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Green Security: A Systematic Review on the Link Between AI-Driven Cybersecurity and Sustainability

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ABSTRACT

The convergence of Artificial Intelligence (AI) and cybersecurity has significantly strengthened the protection of digital infrastructures, particularly in critical domains such as the Internet of Things (IoT), smart grids, and industrial control systems. However, the environmental implications of AI-driven security solutions remain underexplored, despite the growing imperative for sustainability in technology development. This paper presents a systematic review of contemporary research at the intersection of AI, cybersecurity, and sustainability, providing a structured classification of methodologies, application domains, sustainability objectives, and open challenges. Key challenges include the lack of standardized sustainability metrics in cybersecurity, the high energy demands of advanced security models, and the limited adoption of cross-layer and lifecycle-aware optimization techniques. Beyond technical analysis, the review contextualizes its findings within the United Nations Sustainable Development Goals (SDGs), particularly those related to responsible innovation, clean energy, and resilient infrastructure. The findings underscore the need for multi-objective optimization frameworks, lifecycle carbon accounting, and policy-driven strategies that collectively enhance cybersecurity robustness, privacy preservation, and environmental stewardship. This systematic review ultimately establishes a research agenda for the development of AI-based cybersecurity solutions that are not only secure and efficient but also environmentally responsible and aligned with global sustainability objectives.

1. Introduction

Artificial Intelligence (AI) has become a foundational pillar of modern cybersecurity architectures, particularly within critical and interconnected infrastructures such as energy systems, water networks, smart cities, and industrial control environments [1–4]. As digital systems expand in scale, complexity, and autonomy, AI-driven techniques, most significantly Machine Learning (ML) and Deep Learning (DL), have been increasingly adopted to enhance automation, scalability, and resilience in cybersecurity operations [5, 6]. These techniques enable effective intrusion detection, anomaly identification, threat classification, and predictive defense, thereby improving responsiveness to rapidly evolving attack surfaces [7–9]. In parallel, complementary technologies such as blockchain have been integrated into cybersecurity frameworks to support decentralized trust, tamper-resistant logging, and secure data exchange across distributed infrastructures [10, 11].

Simultaneously, digital infrastructures are undergoing a global transition toward sustainability, driven by climate commitments, resource constraints, and the growing demand for long-term operational efficiency [12, 13]. Energy systems, communication networks, and computing platforms are increasingly designed to minimize carbon emissions, reduce energy consumption, and operate under strict resource budgets. Consequently, cybersecurity mechanisms deployed within these environments must satisfy not only performance and reliability requirements, but also environmental and resource-efficiency objectives.

ORCID(s):

Despite their demonstrated effectiveness, AI-driven cybersecurity solutions introduce substantial sustainability challenges. ML and DL-based intrusion detection systems often rely on continuous model training, large-scale data processing, and real-time inference, leading to elevated energy consumption and intensified hardware utilization, particularly in edge computing and Internet of Things (IoT) networks [14–16]. Moreover, complex DL architectures such as convolutional and recurrent neural networks impose latency and memory overheads that conflict with the operational constraints of mission-critical and energy-sensitive infrastructures [17, 18]. Blockchain-based security mechanisms, while effective in strengthening trust and auditability, further amplify sustainability concerns due to the computational intensity of cryptographic operations and consensus processes [19]. Collectively, these characteristics reveal an emerging tension between cybersecurity robustness and system operational sustainability.

Nevertheless, the majority of AI-based cybersecurity research continues to evaluate system effectiveness primarily through conventional performance indicators, such as detection accuracy, false-positive rates, and robustness against adversarial attacks [20, 21]. Environmental and operational dimensions, such as energy consumption, inference latency, lifecycle carbon emissions, and hardware efficiency, are frequently overlooked or treated as secondary considerations. Only a limited number of studies explicitly quantify energy usage or carbon impact in cybersecurity evaluations, while even fewer investigate the trade-offs between security effectiveness and sustainability constraints during deployment [22, 23]. As a result, many AI-driven security solutions that perform well in controlled experimental settings face

significant barriers to adoption in real-world, resource-constrained, and environmentally regulated infrastructures.

This situation exposes a fundamental gap in the existing literature: the lack of an integrated, systematic understanding of how AI-driven cybersecurity mechanisms can be designed, evaluated, and deployed in alignment with sustainability principles. Current research largely isolates cybersecurity objectives from environmental considerations, leading to solutions that are effective at detection but inefficient in terms of energy use, carbon footprint, and long-term operational viability. Consequently, system designers and stakeholders lack clear guidance on how to balance cybersecurity performance with sustainability requirements, limiting the practical deployment of AI-based security solutions in modern critical infrastructures. Motivated by this gap, the present study conducts a systematic review that explicitly bridges the areas of AI-driven cybersecurity and sustainability. Rather than treating sustainability as an auxiliary concern, this review jointly examines the effectiveness of security, energy efficiency, resource utilization, and environmental impact across a broad range of AI-based cybersecurity approaches. To support consistent comparison across studies, a unified set of evaluation dimensions is adopted, including energy consumption measured in Joules per inference, carbon emissions quantified as CO_2 equivalents across lifecycle stages, computational efficiency expressed as operations per Watt, latency–performance trade-offs, and privacy preservation under resource constraints. This integrated perspective enables the identification of recurring limitations, structural trade-offs, and emerging design patterns that characterize sustainability-aware cybersecurity systems.

Existing surveys typically focus on AI applications in cybersecurity or on AI techniques for sustainable computing, rarely addressing both dimensions simultaneously. Several reviews provide detailed taxonomies of AI-based cybersecurity methods while neglecting environmental impact and deployment feasibility in constrained settings [24–26]. In contrast, studies focused on AI for sustainability emphasize energy optimization and emissions reduction, but often overlook adversarial threats, system resilience, and trust requirements [27]. This separation impedes the development of holistic frameworks to guide environmentally responsible cybersecurity design.

To address this limitation, this work systematically reviews peer-reviewed literature published between 2012 and 2025 on AI-based cybersecurity approaches, including ML, DL, federated learning, and blockchain, as applied to sustainable and resource-aware infrastructures. The review examines how existing methods balance security objectives and sustainability constraints, identifies unresolved challenges and contradictions, and synthesizes insights across technical, environmental, and policy dimensions. Furthermore, the study aligns its analysis with key United Nations Sustainable Development Goals (SDGs), notably SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation, and Infrastructure), SDG 12 (Responsible Consumption and Production), and SDG 16 (Peace, Justice, and Strong Institutions) [28]. By integrating cybersecurity performance with sustainability considerations, this review establishes a comprehensive foundation for developing secure, efficient, and environmentally aligned AI-driven cybersecurity systems. Accordingly, this paper makes the following key contributions:

- It provides a systematic review of AI-driven cybersecurity solutions for sustainable, resource-aware infrastructures, covering 13 years of peer-reviewed research.
- It analyzes the technical, environmental, and operational challenges of deploying AI-based security systems under energy and resource constraints.
- It identifies critical gaps in existing literature, including the lack of sustainability-aware security design, the absence of unified evaluation metrics, and the limited availability of real-world validation studies.
- It outlines a research agenda that promotes cross-layer optimization, lifecycle-aware evaluation, and policy-aligned governance mechanisms for sustainable cybersecurity.

The remainder of this paper is organized as follows. Section 2 presents the methodology. Section 2.7 reports the main findings. Section 3 analyzes AI-based cybersecurity approaches, while Section 5 examines their integration with sustainability objectives. Section 6 summarizes the datasets used. Sections 7 and 8 discuss governance considerations and standardization needs. Section 5.4 presents performance–sustainability trade-offs. Sections 11–13 provide the discussion, limitations, research challenges, and concluding remarks.

2. Methodology

This systematic review followed the structured methodology for conducting systematic literature reviews outlined by Kitchenham and Charters [29], ensuring transparency in planning, execution, and reporting. The process began with the formulation of targeted research questions to explore the intersection of AI-based cybersecurity techniques (e.g., ML) and blockchain, and their alignment with sustainability objectives. A list of keywords was then developed to support a systematic literature search strategy across major academic databases. To ensure the relevance and quality of included studies, explicit inclusion and exclusion criteria were established, focusing on works published between 2012 and 2025 that addressed cybersecurity applications within sustainable energy, infrastructure, and resource-constrained environments. A structured data extraction method was applied to capture information pertinent to the research questions, including technical characteristics, evaluation metrics, deployment contexts, and sustainability considerations. Additionally, the synthesis phase involved a thematic analysis of the extracted data to identify current trends, key challenges, and research gaps, particularly the lack of integrated

frameworks that jointly optimize cybersecurity performance and environmental sustainability in AI-driven systems.

2.1. Literature Classification Scheme

To ensure clarity and systematic coverage, the selected studies were categorized along three primary dimensions: (i) AI approach, distinguishing betweenML (e.g., supervised, unsupervised, RL, DL), and blockchain; (ii) sustainability aspect, including energy efficiency, carbon footprint reduction, resource optimization, and privacy preservation; and (iii) application domain,e.g., smart grids, green IoT, edge/fog computing, healthcare, and transportation. This multidimensional classification enables structured analysis, allowing each work to be positioned according to both its technical approach and its sustainability objectives, and facilitating cross-domain comparison.

2.2. Research Questions (RQs)

The primary objective of this systematic review is to investigate how AI techniques and blockchain have been applied to cybersecurity within sustainable infrastructure systems. This includes assessing the current research landscape, evaluating the extent to which sustainability concerns are addressed, and identifying key research gaps that hinder the co-optimization of cybersecurity and environmental objectives. To systematically explore this objective, the following RQs have been formulated:

- **RQ1: What is the current state-of-the-art in applying AI-based cybersecurity methods (ML and blockchain) within sustainable infrastructure domains?**
  This question examines the range of AI-based cybersecurity approaches (e.g., ML) and blockchain that have been applied in sustainable infrastructure contexts, such as smart grids, green IoT, and edge and fog computing. It involves systematically identifying, categorizing, and comparing state-of-the-art techniques, models, and architectures, with a focus on their performance, scalability, and integration with sustainability considerations, including energy efficiency, carbon footprint, and operation in resource-constrained environments.

- **RQ2: To what extent do existing AI-based cybersecurity approaches incorporate sustainability objectives such as energy efficiency, resource constraints, and environmental impact?**
  This question aims to determine the degree to which sustainability considerations are addressed in AI-based cybersecurity research. It focuses on identifying whether these approaches explicitly account for factors such as energy efficiency, operation under resource constraints (e.g., bandwidth, memory, processing power), and environmental impact (e.g., carbon footprint) in their design, deployment, and evaluation.

- **RQ3: What are the key challenges and research gaps in aligning cybersecurity performance with sustainability in AI-driven systems?**
  This question highlights the limitations in current research, including the lack of unified frameworks, the absence of joint evaluation metrics, and the underrepresentation of real-world case studies. It also aims to highlight opportunities to advance co-design strategies and to outline future directions for secure and sustainable infrastructure systems.

2.3. Search strategy

This systematic review employed both automated and manual search strategies to identify a body of literature aligned with the study's objectives. Moreover, computerized searches were conducted across multiple academic databases to ensure broad coverage of relevant research in AI, cybersecurity, and sustainability. Additionally, manual searches were conducted through citation tracking and reference mining of key publications to capture pertinent studies that were not retrieved through keyword-based database queries. The search process adhered to the systematic review guidelines by Kitchenham and Charters [29] and consisted of the following stages:

- A preliminary scoping review was conducted to identify the core thematic intersections: AI-based cybersecurity (e.g., ML) and blockchain, sustainability (encompassing energy efficiency and environmental impact), and deployment contexts (e.g., IoT, edge computing, and smart grids). Based on this, a list of keywords was developed.

- The search strategy was expanded using synonyms, acronyms, and variant terminologies to maximize both precision and recall. Keywords were selected to capture both technical methodologies and application environments across security and sustainability domains.

- The final Boolean query combined three major thematic clusters, infrastructure context, security focus, and sustainability objective, as follows:

```
("Internet of Things" OR IoT OR "cyber-physical
  system" OR CPS OR "edge computing" OR "fog
                  computing"
  OR "cloud computing" OR "wireless sensor
  network" OR WSN OR "5G" OR "6G" OR "embedded
                  system")
                    AND
  ("cybersecurity" OR "cyber security" OR
  "information security" OR "data protection"
  OR "network security" OR "resilience" OR
                  "privacy")
                    AND
  ("sustainability" OR "green ICT" OR "energy
      efficiency" OR "carbon footprint"
        OR "eco-design" OR "ESG")
```

This formulation ensured coverage of research situated at the intersection of secure, intelligent systems and sustainable digital infrastructure. To retrieve relevant papers, we queried three major databases, Web of Science, Compendex, and Inspec, chosen for their interdisciplinary scope and strong indexing in engineering, computer science, and sustainability research[30] [31].[1] To complement this, we conducted a manual search in Google Scholar, screening the first 10 results and using snowballing from key studies. Our search was restricted to peer-reviewed literature published between 2012 and 2025, with particular attention paid to empirical studies, frameworks, and evaluation metrics relevant to both cybersecurity and sustainability. This process allowed us to confirm the novelty of our research scope and ensure a literature base for further screening (see Section 2.3.1).

2.3.1. Inclusion and Exclusion Criteria

To ensure a review, all candidate studies were manually screened by two independent reviewers (the authors). Each reviewer evaluated the titles, abstracts, and, where necessary, full texts of the retrieved papers to assess their relevance to the study's objectives. The selection process was guided by predefined inclusion and exclusion criteria designed to retain high-quality, original contributions at the intersection of AI-based cybersecurity and sustainability. The inclusion criteria were as follows:

- **Scope of Publication:** Both peer-reviewed journal papers and high-quality, peer-reviewed conference papers were considered. This inclusion broadens the literature base while ensuring scientific validity. Conference papers were included if they presented substantial original contributions and were published in reputable venues (e.g., IEEE, ACM, Springer, and Elsevier).
- **Relevance to AI-based Cybersecurity:** The paper must apply AI techniques(e.g., ML) and blockchain to cybersecurity-related problems. This includes, but is not limited to, IDS, anomaly detection, threat modeling, data privacy, and secure communication systems.
- **Context of Sustainability or Resource Constraints:** The study must be applied in a sustainability or resource-constrained context, e.g., smart grids, energy-aware computing, edge/fog environments, renewable-powered infrastructures, systems operating under energy and hardware constraints.
- **Publication Timeframe:** Studies published between 2012 and 2025 were included to reflect developments in both cybersecurity and sustainability over the past thirteen years.
- **Language:** Only papers written in English were included to maintain consistency and ensure accessibility.

[1] Web of Science: https://www.webofscience.com; Compendex and Inspec: https://www.engineeringvillage.com;

To maintain the quality and relevance of the systematic review, papers were excluded based on the following criteria:

- **Irrelevant Scope:** Studies that did not involve both AI-based cybersecurity and a sustainability and resource-constrained context were excluded. For example, papers on AI for sustainability that lacked a security component or security solutions without AI or sustainability elements were removed.
- **Publication Type:** The following were excluded: review papers, surveys, editorials, opinion pieces, short abstracts, tutorials, workshop summaries, white papers, technical reports, and academic theses. These types of documents often lack detailed methodological content or original empirical results.
- **Non-English Publications:** Papers not written in English were excluded.
- **Duplicate Studies:** If the same research appeared in multiple formats (e.g., a conference and later an extended journal version), only the most complete and recent version was retained.

2.4. Paper Selection

This section details the procedure followed for selecting relevant studies and filtering out those that did not meet the eligibility criteria, following the PRISMA 2020 guidelines [32, 33]. As shown in Figure 1, the initial search process retrieved a total of 8,000 records from three major databases: 3,500 from Compendex, 2,800 from Inspec, and 1,700 from Web of Science. Duplicate records were identified and removed using the Rayyan systematic-review tool [34], resulting in the elimination of 5,200 duplicates and leaving 2,800 unique records for screening.

2.4.1. Duplicated Papers

After duplicate removal, the remaining 2,800 records were screened by title and abstract to assess their relevance to the study objectives. At this stage, 2,500 records were excluded because they did not meet the fundamental criteria for AI-based cybersecurity and sustainability. Consequently, 300 reports were shortlisted for full-text retrieval and in-depth assessment. Of these, 12 reports could not be retrieved due to language or access restrictions, leaving 288 reports for detailed evaluation against the predefined inclusion and exclusion criteria (see Section 2.3.1).

From this set, 244 reports were excluded for the following reasons: 65 were unrelated to the target domain (e.g., healthcare or natural-language processing); 39 did not employ AI-based techniques; 34 lacked sustainability or resource-awareness components; 26 were purely theoretical or conceptual without empirical validation; 20 were duplicate or overlapping publication versions; 18 were off-topic after full-text inspection; 16 lacked sufficient technical detail; 14 were of low experimental quality; 7 did not primarily address cybersecurity; and 5 were unavailable in English or inaccessible.

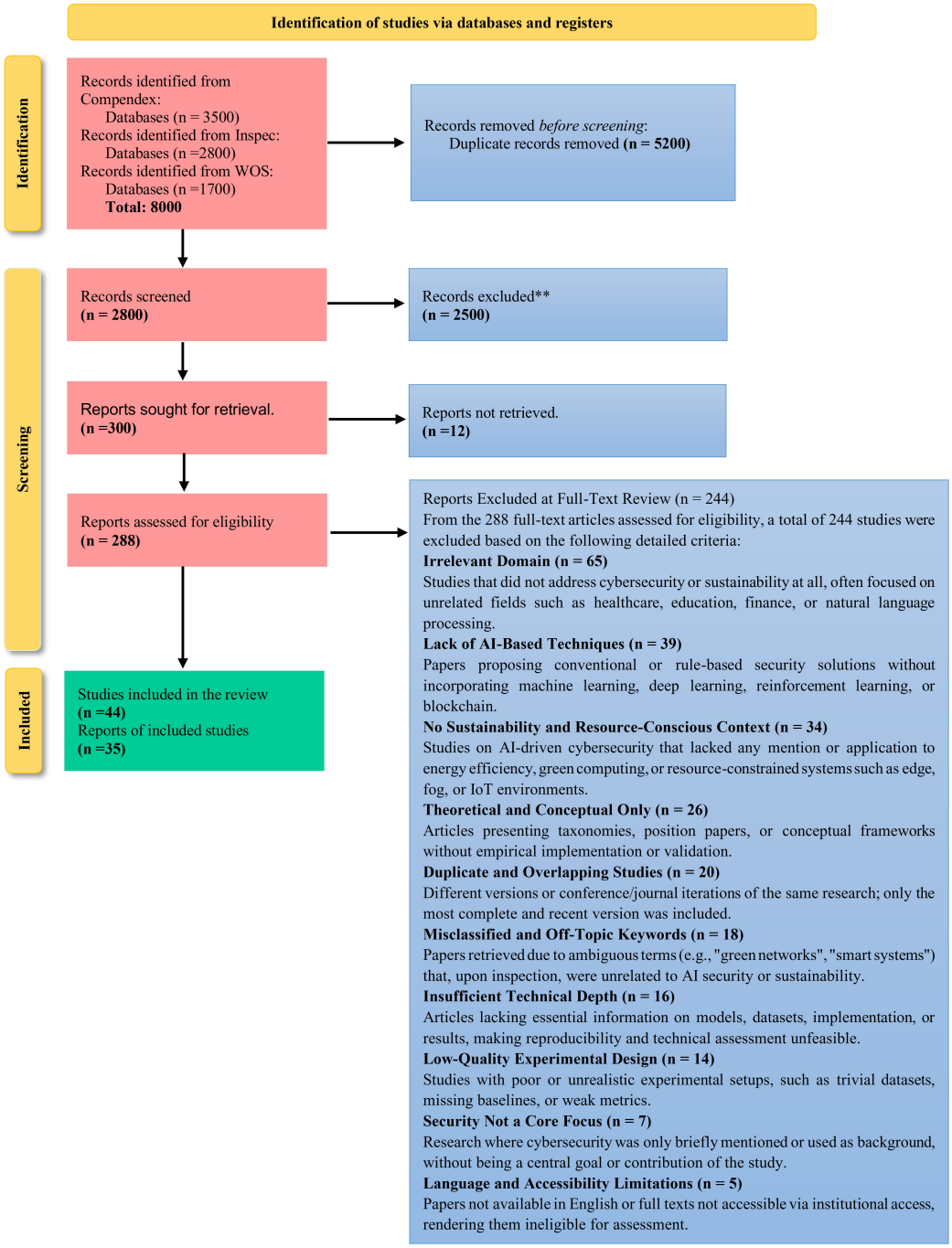


Figure 1: PRISMA 2020 flow diagram showing the study-selection process. A total of 44 studies (35 distinct reports) were included in the final synthesis.

As a result, 44 studies were included in the final systematic review, corresponding to 35 distinct research reports, as illustrated in Figure 1. This distinction accounts for cases in which the same underlying study appeared in multiple publication formats (e.g., an initial conference paper followed by an extended journal version). In such cases, all reports were considered during screening, while only the most complete and recent version was treated as a distinct report for synthesis and comparative analysis. The included studies form the foundation of the analysis presented in the subsequent sections and represent the state of the art in applying AI techniques, particularly ML, FL, and blockchain, to cybersecurity challenges in sustainable and resource-constrained environments.

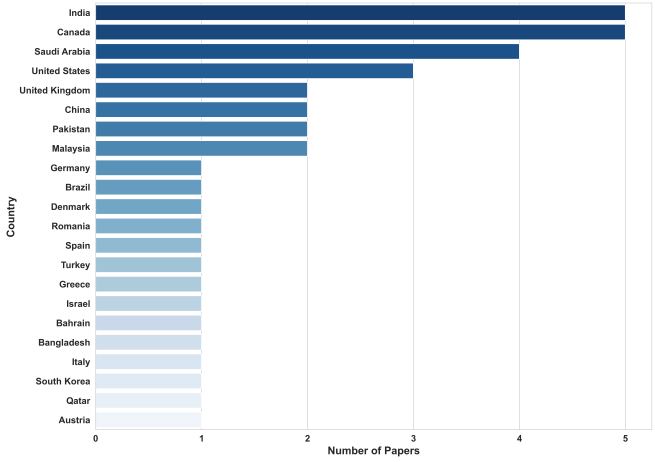


Figure 2: Geographical distribution of research publications by country.

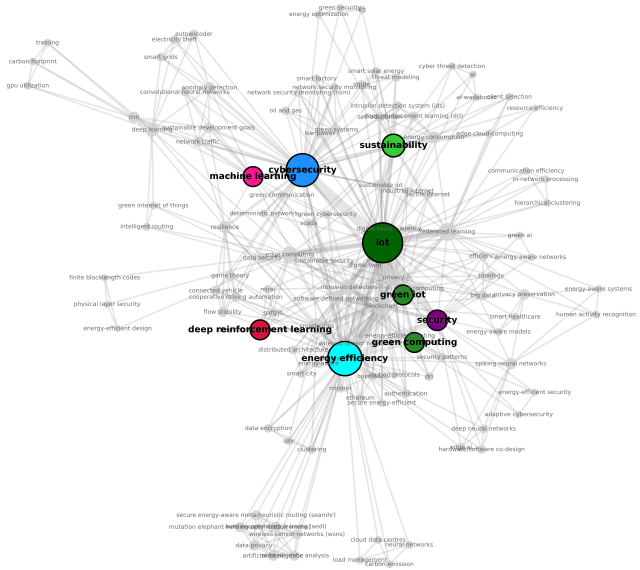


Figure 3: Keyword co-occurrence network based on titles, abstracts, and author keywords.

2.5. Analysis of Included Papers

Figure 2 illustrates the geographical distribution of research contributions in the domain of AI-based cybersecurity applied to sustainable infrastructures. This data, derived from the final set of included studies, reveals that Canada, India, and Saudi Arabia each produced the highest number of publications (5papers), indicating significant research leadership in this emerging and interdisciplinary field. The United States and the United Kingdom follow closely, reflecting robust academic ecosystems and institutional support for integrating artificial intelligence, cybersecurity, and environmental sustainability. Countries such as China, Pakistan, and Malaysia demonstrate moderate representation, likely impacted by increasing national investments in digital transformation and green technology initiatives. Similarly, European countries such as Italy, Germany, and Denmark show consistent engagement, often driven by regulatory frameworks and strategic alignment with long-term sustainability goals. This distribution also reflects broader international commitments to the United Nations SDGs, particularly SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation and Infrastructure), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action). The convergence of AI, security, and sustainability closely aligns with these global objectives, particularly in contexts such as smart grids, low-power IoT, and green digital infrastructures. Specially, several countries (e.g., Austria, Greece, and South Korea) are represented by only one publication each, indicating isolated but potentially high-impact research efforts. The landscape suggests that while global interest in this field is expanding, contributions remain unevenly distributed. This highlights both the leadership of key research hubs and the untapped potential for broader international collaboration, particularly in low- and middle-income regions, where secure, sustainable digital infrastructure is crucial for equitable development.

Figure 3 illustrates the co-occurrence network of the most frequently appearing keywords across the reviewed literature, extracted from titles, abstracts, and keyword sections. The network reveals well-defined thematic clusters that characterize the evolving research landscape at the intersection of AI-based cybersecurity and sustainability. Central nodes such as *cybersecurity*, *machine learning*, *sustainability*, *energy efficiency*, *green computing*, and *deep reinforcement learning* exhibit high connectivity, underscoring their conceptual prominence within the field. The spatial proximity of terms reflects topical convergence; for example, *green IoT*, *smart grids*, and *secure edge computing* are situated near both AI and sustainability nodes, indicating a growing emphasis on energy-aware security architectures. Node size corresponds to keyword frequency, while edge thickness denotes the strength of co-occurrence between terms. Additionally, the network highlights the field's interdisciplinary nature and underscores the emerging integration of AI-driven security solutions with environmental and operational priorities. It also exposes dominant research themes alongside underexplored intersections that offer valuable directions for future investigation.

2.6. Data Extraction

We extracted both general information and specific details relevant to the research questions formulated for this systematic review. This structured approach ensured a consistent, systematic evaluation of each selected paper's contribution to AI-based cybersecurity in sustainable infrastructure contexts. The following general information was collected from each paper:

- Title
- URL to the paper
- Authors

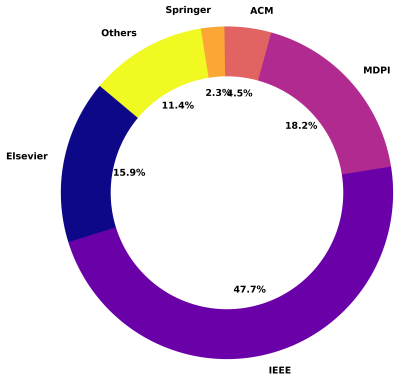


Figure 4: Distribution of selected papers by journal.

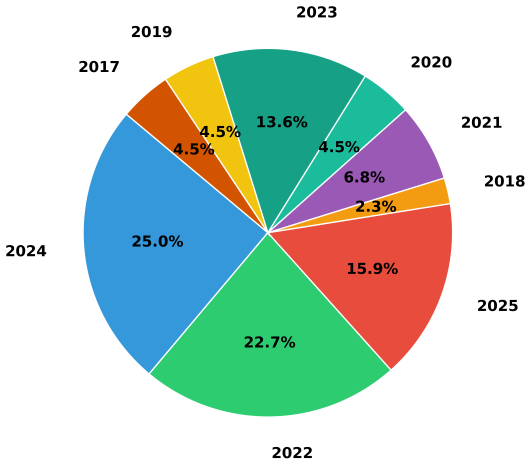


Figure 5: Annual trend of research paper publication.

- Year of publication
- Publication venue

2.7. Statistical Findings

Figure 4 presents the selected studies across major academic publishers. IEEE accounts for the largest share, with 47.7% of all reviewed papers, underscoring its prominent role in advancing research at the intersection of AI, cybersecurity, and sustainable systems. MDPI follows with 18.2%, likely reflecting the publisher's recent focus on interdisciplinary topics and open-access dissemination. Elsevier contributes 15.9%, maintaining its reputation for publishing high-impact research in engineering and computing domains. Other publishers, including ACM (4.5%), Springer (2.3%), and a variety of smaller outlets grouped under "Others" (11.4%), indicate a growing diversity in publication venues. This distribution highlights not only the central publication channels for this research domain but also signals increasing interest from journals that traditionally serve applied computing, sustainability science, and emerging technologies.

Figure 5 illustrates the temporal trend in publication volume from 2017 to 2025. A marked increase in research output is observed from 2022 onward. The years 2022 and 2024 were particularly productive, accounting for 25.0% and 22.7% of the reviewed studies, respectively. This surge likely corresponds with the rising global emphasis on digital transformation and sustainable infrastructure, in alignment with international policy frameworks such as the United Nations' SDGs, particularly SDG 9 (Industry, Innovation, and Infrastructure) and SDG 13 (Climate Action). Specifically, 2025 already shows strong representation (15.9%), indicating sustained momentum and the ongoing relevance of this research theme. Earlier years, e.g., 2017–2020, contributed fewer papers, reflecting the field's relatively recent emergence. The growth trend suggests that integrating AI-driven cybersecurity into sustainability-critical systems has matured into a priority research area, likely driven by the convergence of innovative technologies, green computing, and the need for resilience in digital infrastructures.

3. Categorization of state-of-the-art papers

To address RQ1, this section presents a structured categorization of the reviewed state-of-the-art papers that integrate AI techniques and blockchain with cybersecurity in the context of sustainability. The categorization is organized based on methodologies, application domains, and sustainability objectives, providing an overview of current research trends and technical approaches in this interdisciplinary field. This section synthesizes the state of the art by organizing prior research into thematic clusters that reflect the main sustainability–cybersecurity trade-offs. Rather than listing each study sequentially, we highlight cross-cutting trends, comparative findings, and emerging design principles. Specifically, studies are grouped by methodological orientation, federated and edge learning, blockchain integration, ML-based IDS, green IoT, and optimized networking, each addressing distinct balances between computation, communication, and energy efficiency. The accompanying tables (1, 3, 4, 5) summarize these themes to reinforce the narrative synthesis.

3.1. Federated Learning (FL) and Edge Computing

Savoia et al. [35] present Eco-FL, a novel framework that enhances energy efficiency in FL within edge cloud computing by optimizing client selection. The method employs an optimization model that balances two criteria: clients' residual energy and data entropy, ensuring that energy-depleted devices are excluded while maximizing informational contribution. The approach was implemented using the Flower framework and evaluated on CIFAR-10, CIFAR-100, and CINIC-10 datasets under heterogeneous and homogeneous energy scenarios. The LeNet model was trained with SGD for 1500 rounds and up to 100 clients. Eco-FL was benchmarked against FedEntropy, FedAvg, and FedProx. Results showed that Eco-FL reduced energy consumption significantly; for example, on CIFAR-10 with $\beta = 0.1$, Eco-FL consumed 227 Wh compared to FedAvg's 6,923 Wh. Similarly, on CIFAR-100, energy consumption dropped

to 1524 Wh for Eco-FL versus 29,714 Wh for FedAvg. In heterogeneous energy scenarios, Eco-FL achieved even lower consumption, e.g., for CIFAR-10 ($\beta = 0.1$), 125 Wh. Accuracy remained competitive, efficiently reaching the target threshold (40%) while ensuring battery-aware client utilization. Sensitivity analysis of the α parameter showed that assigning partial weight to entropy, alongside energy, optimally balances consumption and accuracy. Eco-FL maintained robust performance even under simulated device failures, demonstrating its suitability for real-world, resource-constrained environments.

Qiu et al. [36] present the first systematic study of the carbon footprint of FL, proposing an analytical model to estimate CO_2 emissions that accounts for both hardware energy consumption and communication costs. The authors conducted extensive experiments on FL hardware (NVIDIA TX2 and NX) using datasets, e.g, CIFAR-10, ImageNet, FEMNIST, Speech Commands, and CV Italian, under various FL strategies (FedAvg and FedAdam) and data distributions (IID and non-IID). Results reveal that FL can emit up to two orders of magnitude more carbon than centralized training, particularly in non-IID settings with high communication overhead. For example, CIFAR10 non-IID with one local epoch resulted in over 730g CO_2 versus 3g in centralized setups (Australia, PUE 1.67). However, in lightweight tasks such as Speech Commands and CV Italian, FL occasionally yields lower emissions than centralized methods. The study also highlights the impact of location-specific carbon intensity, local epochs, aggregation strategies, and hardware efficiency. Communication contributed between 0.7% (ImageNet) and 96% (CIFAR-10) of total emissions.

Baqer et al.[37] introduce the Tiny Federated Nearest Neighbor (TFNN) approach to achieve energy-efficient FL in IoT networks. By combining in-network processing with hierarchical clustering using the LEACH protocol, TFNN applies K-Nearest Neighbor (KNN) locally at IoT nodes. Cluster heads aggregate local classifications and transmit the results to the base station, enabling single-round pattern recognition and avoiding the high communication overhead associated with traditional FL methods. Experiments using the MNIST dataset on a simulated 50-node IoT network demonstrated that TFNN nearly doubled network lifetime compared to direct communication approaches, maintaining stable accuracy across 800 rounds. Energy consumption was significantly reduced by limiting long-range transmissions, and communication delays were minimized via single-round decision-making. In contrast to FedAvg, FedProx, and FedBuff, which require approximately 231, 228, and 31.9 rounds, respectively, for convergence, TFNN achieves efficient pattern recognition in a single round.

Aljrees et al.[38] introduce a green and sustainable FL platform to bolster IoT security while reducing communication costs and energy consumption. The framework synergistically integrates FL, the Quondam Signature Algorithm (QSA), and efficient encryption mechanisms to protect against evolving threats, including man-in-the-middle attacks. The QSA employs timestamp-based matrix encryption with a substitution box to generate one-time, unique signatures, thereby enhancing device authentication. FL facilitates decentralized model training across IoT devices, preserving data privacy and enabling continuous learning. The proposed system utilizes an elliptic curve digital signature-based online/offline scheme, thereby optimizing both time complexity and communication efficiency. It especially outperforms the Slow Block Move (SBM) approach, reducing both the number of rounds and the number of signature operations.

Khan et al. [39] present a privacy- and energy-aware FL framework for Human Activity Recognition (HAR) using wearable sensors that employs a hybrid Spiking Neural Network and Long Short-Term Memory (S-LSTM) model. The S-LSTM combines the event-driven efficiency of SNNs with the sequential data processing capabilities of LSTMs, trained via surrogate gradient learning and backpropagation through time. Evaluations on the UCI and Real-World HAR datasets achieved accuracies of 97.36% and 89.69%, respectively, outperforming LSTM, CNN, and spiking-CNN (S-CNN) models. The S-LSTM improved energy efficiency by 32.3% compared to LSTM. The framework supports client personalization through local model fine-tuning, increasing accuracy by up to 9% per participant. Additionally, random client selection during FL rounds further enhanced energy efficiency, resulting in a 24.41% reduction in consumption compared to LSTM.

Adeb Salh et al. [40] introduce a sustainable resource allocation framework for IoT networks that utilizes FL-enabled digital twins in conjunction with blockchain and DRL. The proposed Networks with Digital Twin Edges (NDITE) architecture enables secure, energy-efficient real-time data processing at the edge. FL enables privacy-preserving decentralized model training, while blockchain ensures data integrity and security. A DRL agent, utilizing a Deep Neural Network (DNN), dynamically optimizes bandwidth allocation and user scheduling. Simulation results demonstrate that the DRL-based DT framework achieves local execution of 47.5% of computing tasks with only 1 MHz bandwidth, surpassing the 43% rate of traditional Resource Allocation (RA) methods. The system also reduces energy consumption and system latency, while enhancing inference accuracy and cumulative rewards. This integration of DT, FL, and DRL provides an effective solution for sustainable, low-latency, and energy-efficient IoT networks in beyond 5G scenarios.

Across these studies, a consistent trend emerges: FL provides strong privacy preservation and distributed security while incurring communication and synchronization overhead. Most frameworks aim to minimize energy use through client selection, adaptive scheduling, and lightweight aggregation; however, carbon footprint analysis remains rare. This highlights a growing shift from purely accuracy-driven FL toward sustainability-aware FL, balancing computation, communication, and emissions.

3.2. Blockchain-based Security and Energy Efficiency

In line with the systematic review's focus on AI-based cybersecurity in sustainable infrastructure domains, this subsection surveys blockchain-based security approaches specifically designed for resource-constrained, sustainability-focused contexts, such as IoT-based systems, WSNs, and smart cities.

Gurram et al. [41] propose the Secure Energy Aware Meta-Heuristic Routing (SEAMHR) protocol for IoT-based Wireless Sensor Networks (WSNs) to address security, energy efficiency, and data reliability challenges. The protocol employs Mutation Elephant Herding Optimization (MEHO) for routing, leveraging aggregated residual energy, hop count, and link integrity to select optimal next-hop nodes. Additionally, SEAMHR integrates counter mode cryptography with AE (CTR-AE) for secure data encryption and decryption. Simulation results in MATLAB show that SEAMHR improves network throughput by 6.5% to 17.75% and reduces the packet drop ratio by 5.98% to 18.32% compared to SEHR, HBEER, and SecTrust-RPL. It also achieves an average 35.80% reduction in end-to-end delay, a 52.73% reduction in energy consumption, and a 68.29% reduction in computational overhead. Faulty routes are reduced by 54.15%, and network overhead is minimized by 68.62%.

Hamad Aldawsari et al. [42] propose a blockchain-based framework for secure and energy-efficient IoT-based WSNs in smart cities. The framework integrates blockchain for decentralized and tamper-proof spectrum management with cognitive radio to enhance spectrum efficiency. A smart clustering mechanism, combined with trust-based spectrum sensing, optimizes energy consumption and prolongs the network lifetime. The proposed methodology incorporates an energy-aware node selection algorithm based on trust values, along with a return function that accounts for both energy consumption and sensing performance. The system was evaluated using MATLAB (R2020a), showing improvements in key metrics: the false alarm probability was reduced to 0.68, compared with 0.69 and 0.70 for the existing FOF and TTA methods, respectively. Spectrum sensing performance gain reached 0.92, surpassing FOF (0.72) and TTA (0.33). The missed-detection probability was significantly reduced to 0.14, compared with 0.78 (FOF) and 0.99 (TTA). The architecture also demonstrated efficient energy trading among sensor nodes and minimized communication overhead.

Ahmed et al. [43] propose an Energy-Efficient Data Aggregation Mechanism (EEDAM) for IoT networks secured by blockchain. The architecture leverages edge computing and fuzzy clustering to reduce data redundancy, optimize energy consumption, and enhance security in IoT networks. The system organizes sensor nodes into clusters using a fuzzy similarity matrix, selects redundant nodes for sleep scheduling, and efficiently aggregates data. Blockchain is integrated at the cloud layer to validate edge servers and ensure secure data transactions via proof-of-authority. Simulations on networks with up to 1000 nodes demonstrate that EEDAM reduces energy consumption by up to 35% compared to the EEHS and ESSM protocols, with data aggregation accuracy exceeding 90%. Moreover, time delays are minimized, especially in dense deployments, and gas consumption for blockchain operations scales linearly with the number of service requests. EEDAM also enhances data security, prevents malicious edge devices, and maintains high scalability with minimal overhead, positioning it as a practical framework for secure and sustainable IoT systems.

Mohammed [44] introduces an optimized and secure energy-efficient blockchain-based framework for IoT by integrating Software-Defined Networking (SDN) with blockchain technologies. The proposed SDN-blockchain classifier utilizes a trust-based IDS with graylist and whitelist packet classifiers to detect and mitigate MAC flooding attacks, thereby enhancing both network security and efficiency. Implemented on an OpenStack cloud platform utilizing OpenDaylight SDN controllers and Ethereum, the system optimizes performance metrics, including throughput, energy consumption, end-to-end delay, and CPU usage. Experimental results show significant improvements, with energy consumption reduced by approximately 50% compared to the BCF, AS-CIR, and BSDNSR methods. Packet loss recovery was improved by 23%, and end-to-end delay and CPU usage were minimized under attack scenarios.

Rehman, A. et al. [45] proposes a blockchain-based framework to enhance the security and energy efficiency of WSNs. The authors address critical IoT security threats, including node capture, replay, side-channel attacks, and false data injection. The solution combines blockchain's decentralized ledger with a clustering strategy (the LEACH protocol), in which cluster heads aggregate data to reduce energy consumption. Cryptographic techniques such as Merkle trees and SHA-256 hashing ensure data integrity and authentication. The experimental results indicate that the probability of data manipulation in blockchain-enabled WSNs is reduced by approximately 60% compared to traditional WSNs. Furthermore, the blockchain system's data-processing capabilities remain competitive, with only a minor reduction in throughput compared to conventional methods. A comparative analysis also revealed that message tampering is significantly less likely in blockchain-based systems, with data integrity consistently maintained, even when up to 87 blocks were evaluated for tampering scenarios.

Goyat, R. et al. [46] propose BENIGREEN, a blockchain-based energy-efficient and privacy-preserving data management scheme tailored for green IoT in smart cities. The framework addresses decentralization, energy conservation, and privacy. It achieves a 45.31% reduction in computational latency, a 31.21% improvement in throughput, and a 39.07% reduction in residual energy, communication cost, link quality, and connectivity. The authentication scheme integrates certificate issuance, revocation, and dynamic key management. Experimental results show that BENIGREEN achieves 45.31% reduction in computational latency, 31.21% improvement in throughput, and 39.07% reduction in energy

consumption compared to existing solutions. Although operation time is 21.2% higher due to blockchain overhead, BENIGREEN outperforms in overall security, achieving resilience against replay, forging, and modification attacks. Formal analysis further confirms the protocol's robustness under the Random Oracle Model (ROM), ensuring privacy, traceability, and revocability.

Islam, M.J. et al. [47] presents a blockchain-SDN-based, energy-aware, and distributed secure architecture for IoT in smart cities, integrating Network Function Virtualization (NFV) to optimize energy consumption and enhance security. The paper proposes an energy-efficient Cluster Head Selection (CHS) algorithm that selects cluster heads based on residual energy and node proximity, significantly reducing energy consumption. Simulation results show that the proposed CHS algorithm outperforms LEACH-Mobile, particularly in reducing power consumption for node counts ranging from 20 to 25. In throughput evaluation, the proposed system achieves higher performance than both Core and DistArch-SCNet models, especially for 2400 transaction requests. Response time is also lower, particularly when the number of requests exceeds 1200. Additionally, gas consumption increases linearly with the transaction count, thereby maintaining efficiency.

Blockchain-based security solutions demonstrate high resilience and transparency; however, their sustainability varies depending on the consensus mechanism employed. Recent research increasingly replaces Proof-of-Work with Proof-of-Stake, DAG, and hybrid schemes to reduce power consumption. This thematic cluster reveals an evolution from energy-intensive trust models to greener, adaptive blockchains aligned with IoT and smart-city sustainability goals.

3.3. ML-based IDS

Jamshidi et al. [23] propose SecuEdge-DRL, a self-adaptive IDRL-based IDS for securing IoT edge devices while optimizing energy consumption. The DRL-based IDS is integrated with the MAPE-K control loop (Monitor, Analyze, Plan, Execute, and Knowledge), enabling real-time detection and adaptive responses to evolving threats, such as DDoS, DoS GoldenEye, DoS Hulk, and Port Scanning. The system was trained using the CICIDS2017 dataset, selecting 23 critical traffic features via a correlation-based feature selection method. A real-world deployment on Raspberry Pi edge gateways demonstrated the system's effectiveness, achieving 92.54% detection accuracy, 95.73% precision, and an F1-score of 93.88%. The model efficiently balances security with resource constraints by minimizing energy consumption, CPU load, and CPU usage under attack scenarios. Adaptive policies also enhance hardware resilience by monitoring CPU temperature, memory usage, and failure risks, ensuring long-term stability in IoT systems.

Jamshidi et al.[48] present EdgeShield-DRL, a DRL-based IDS tailored for IoT edge gateways to defend against Mirai and Gafgyt botnet attacks. The IDS uses a Deep Q-Network (DQN) model, represented as a Markov Decision Process, enabling dynamic threat mitigation while ensuring resource efficiency. Evaluated on the N-BaIoT dataset, the system achieved 97% detection accuracy during training and 96% in real-time scenarios, with F1-scores of 0.97 and 0.96, respectively. A real-world testbed on Raspberry Pi devices demonstrated CPU usage between 60-70%, memory usage of 1520-1600MB, and IDS response times under 0.3 seconds. Energy consumption during attacks rose slightly to 3.5 J with minimal carbon emissions (≈0.0025 Kg). The model effectively balances security, energy efficiency, and environmental sustainability, making it suitable for critical applications, e.g., smart cities and healthcare.

Jamshidi et al. [22] introduce a dynamic security pattern selection framework utilizing DRL to enhance the security and energy efficiency of IoT edge gateways. A DQN-based IDS dynamically selects appropriate IoT security patterns according to the detected attack type, including DDoS, Port Scanning, DoS Hulk, and DoS GoldenEye. Trained on the CICIDS2017 dataset, the system achieved 97% accuracy in training and 95% during real-time testing, with an F1 score of 0.94. Real-world testbed evaluations on Raspberry Pi devices revealed low CPU usage (minimum 0.69%), energy-efficient operation, and rapid IDS response times (1 second).

Maiwada et al. [49] present an energy-efficient IDS for 5G networks that focuses on mitigating DDoS attacks. The paper developed a framework utilizing traffic volume monitoring, statistical analysis, and adaptive system configuration to optimize energy consumption while preserving detection accuracy. The framework was implemented in a NETSIM simulation environment, employing DDoS datasets from Kaggle. Detection rates achieved were 98% via traffic volume analysis, 98% with pattern recognition, and 96% using statistical analysis. The IDS dynamically allocates computational resources, such as CPU, memory, and bandwidth, according to identified risks, thereby reducing energy consumption during normal operations while scaling resources during attacks. Performance metrics, such as throughput, were evaluated over 10-50 ms, ensuring effective DDoS mitigation with minimal impact on network performance. The system includes continuous monitoring for real-world adaptability, offering a resilient and sustainable security solution for 5G infrastructures.

Bushehri et al. [50] propose a DL-based anomaly detection framework for identifying energy overconsumption in green IoT edge networks. Utilizing a public dataset of 18,448 peer-to-peer IoT communication samples, the authors employ a modified 3-layer variational AE with a redesigned loss function that prioritizes the most influential feature, Protocol, determined via linear regression analysis. The model is trained on normal consumption data and then tested on a mixed sample of normal and abnormal observations. The proposed framework achieved an accuracy of 94.8%, precision of 95.2%, and F1-score of 95.0%, significantly outperforming baseline clustering methods and standard VAEs. Furthermore, the model demonstrated a mean absolute error of 0.06014 and a root mean square error of 0.24489. The use of the ReLU activation function, combined with a 3-layer architecture, provided optimal performance with a

manageable computational cost. Designed for on-demand diagnostics, the framework utilizes dynamic binary instrumentation to collect data with minimal overhead, providing real-time feedback to optimize transmission settings and thereby enhancing energy efficiency in IoT networks.

Takiddin et al. [51] propose a deep AE-based framework for detecting electricity theft cyberattacks in smart grids using only benign consumption data. The paper explores three architectures: simple AE, variational AE, and AE with attention, implemented in both fully connected and sequence-to-sequence LSTM structures. Experiments were conducted on two datasets: the State Grid Corporation of China and the Irish Smart Energy Trial, augmented with six types of synthetic attack functions. The AEA model achieved the highest performance, attaining a detection rate of 94% and a false alarm rate of 5%, improving DR by 4-21% and reducing FA by 4-13% compared to state-of-the-art methods. The LSTM-based models demonstrated superior performance by effectively capturing temporal patterns, while the AE's attention mechanism further enhanced the detection of complex theft behaviors. The models also demonstrated robustness against zero-day attacks and provided near-real-time detection with minimal computational cost during online deployment.

Kalra et al. [52] present a CNN-based anomaly detection model designed for green cybersecurity and the development of eco-resilient networks. Using the CICIDS2017 dataset, which includes both regular and malicious network traffic, such as DDoS and brute force attacks, the CNN model was trained and evaluated for its detection capabilities. The architecture comprises convolutional layers for automated feature extraction, pooling layers for dimensionality reduction, and fully connected layers for final classification. The model achieved 93.64% accuracy, with high precision, recall, and F1 scores, demonstrating effective anomaly detection. Training and validation analyses indicated strong learning performance, although some overfitting was observed. The approach also successfully minimized false positives, enhancing its practical applicability. Although the study does not directly quantify energy efficiency, it aligns with SDG 9 (Industry, Innovation, and Infrastructure) and SDG 16 (Peace, Justice, and Strong Institutions) by advancing secure and sustainable digital infrastructures.

Saxena et al. [53] introduce the Sustainable and Secure Load Management (SaS-LM) model, which aims to enhance the sustainability, energy efficiency, and security of cloud data centers (CDCs). The framework employs a Multi-layered Feed-Forward Neural Network (MFNN) optimized with a novel Dual-Phase Black Hole Optimization (DPBHO) algorithm to predict resource usage and detect congestion. For secure and efficient Virtual Machine Placement (VMP), a Multi-objective DPBHO (M-DPBHO) combined with Pareto-based optimization is implemented to minimize energy consumption, carbon footprint, and energy consumption effectiveness, while maximizing resource utilization and reducing security risks. Evaluated on Google cluster trace datasets, the model achieved reductions of 46.9% in carbon emissions and 43.9% in energy consumption, along with a 16.5% improvement in resource utilization. Additionally, the probability of security breaches was decreased by up to 36.4% compared to existing solutions. SaS-LM effectively balances the trade-offs between sustainability and security in green cloud computing environments.

Szymanski [54] presents a green, secure, and energy-efficient architecture for the industrial and tactile IoT by integrating deterministic virtual networks, SDN, and lightweight Layer 2 encryption. Deterministic packet switches under SDN control enable congestion-free, ultra-low-latency communications while inherently detecting intrusions by identifying unscheduled packet transmissions within microseconds. A hardware testbed with 26 FPGA-based switches emulating the US IoT backbone achieved 93% link utilization, queuing delays below 2μs, and reduced buffer sizes by a factor of approximately 1,000,000 compared to best-effort networks. The system supports data rates in the Tbps range, with significant energy and capital cost savings, potentially reducing costs by up to 100 times.

Halabi et al. [55] investigates adaptive cybersecurity as a key enabler for green and sustainable IoT systems, proposing three primary research directions: (i) dynamic threat prevention using moving target defense to shift attack surfaces with minimal resource impact dynamically; (ii) optimized deployment of IDS incorporating RL to produce adaptive, energy-efficient defense policies; and (iii) risk-driven threat mitigation based on continuous security risk assessments, facilitated by game-theoretic models for strategic defense planning. The paper highlights the environmental costs of conventional security mechanisms and argues for integrating AI methods, specifically RL and game theory, into the cybersecurity design for IoT.

Viegas et al. [56] introduce an energy-efficient anomaly-based IDS for embedded systems and compare software and hardware implementations of ML classifiers. The key contributions are: (i) a novel low-power feature extraction algorithm optimized for hardware implementation; (ii) a dual-objective feature selection method utilizing NSGA-II to balance accuracy and energy consumption; and (iii) energy profiling of Decision Tree (DT), Naive Bayes (NB), and KNN in both software and hardware. The hardware feature extractor consumed only 12% of the energy required by a commercial tool, and dual-objective feature selection reduced energy consumption by up to 93%. The hardware implementation of the most efficient setup (feature extractor and DT classifier) consumed just 5.7% of the software energy consumption, with a minor accuracy trade-off of 0.9%. Experiments on a 27 million packet dataset validated these savings, demonstrating the effectiveness of hardware-accelerated ML-based IDS in resource-constrained environments.

Alsulami et al. [57] present an AI-driven model, Artificial Fish Swarm-driven Weight-normalized Adaboost (AF-WAdaBoost), designed to enhance sustainability in cyber threat detection within IoT networks. The model synergizes Artificial Fish Swarm Optimization (AFSO) with a

Weight-normalized AdaBoost ensemble to optimize classifier parameters, thereby enhancing classification performance. Evaluated on CICIDS2017, NSL-KDD, and UNSW-NB15 datasets, the model achieved an accuracy of 98.69%, precision of 95.72%, and F-measure of 94.86%, surpassing traditional methods such as RF, MLP, SVM, and DT. The AF-WAdaBoost approach efficiently adapts to dynamic threats, reduces energy consumption, and minimizes system resource strain, thus promoting sustainability in IoT cybersecurity.

ML-based IDS research emphasizes the trade-off between detection accuracy and resource utilization. RL and DL methods achieve strong precision but remain energy-intensive, whereas lightweight decision-tree and ensemble models offer a balanced trade-off in terms of sustainability. Collectively, these works advance adaptive, self-optimizing IDS that align security robustness with efficiency in the constrained edge gateway.

3.4. Green IoT and Sustainable Smart Cities

Haseeb et al. [58] present an Intelligent and Secure Edge-enabled Computing (ISEC) model for sustainable cities that leverages Green IoT. The model integrates DL for optimal data routing, using a linear activation function that accounts for node density, energy consumption, and link congestion to determine efficient paths. Security is addressed via a symmetric cryptosystem (Diffie-Hellman) for key exchange and a blockchain-based distributed hashing mechanism to ensure data integrity and authentication across multi-hop IoT networks. Simulations conducted in NS-3 with 200 randomly deployed sensors demonstrated that ISEC outperforms existing methods, reducing energy consumption by 21%, increasing network throughput by 15%, lowering end-to-end delay by 12%, decreasing route interruptions by 36%, and minimizing network overhead by 52%. These improvements highlight ISEC's capability to provide energy-efficient, secure, and reliable data transmission for smart, sustainable city environments.

Fraga-Lamas et al. [59] investigate the synergistic integration of green IoT and Edge-AI to enable a sustainable digital transition within the context of the smart circular economy and Industry 5.0. The paper proposes a framework combining IoT, edge computing, and AI to reduce the environmental footprint of digital infrastructure. A practical use case is presented that involves an innovative workshop that leverages mist computing with AI-enabled IoT nodes to enhance operator safety and operational tracking. The study quantifies energy consumption and carbon footprint, showing that edge-AI reduces data transmission and energy costs compared to traditional cloud models. Techniques such as TinyML are discussed for deploying low-power ML models on constrained devices. Although no new AI algorithm has been developed, the paper thoroughly examines AI's critical role in achieving sustainability, energy efficiency, and real-time analytics at the edge.

Brudni et al. [60] introduce the "green security" framework, which offers a taxonomy and methods for measuring and optimizing the energy consumption of cybersecurity solutions. Key inefficiencies, such as redundant endpoint scanning, rescan of unchanged files, and overlapping controls, are identified. The paper develops a process-agnostic energy measurement framework that utilizes resource utilization metrics (CPU, RAM, and I/O) and an ML-based prediction model (RF) to estimate energy consumption across diverse hardware platforms. Several antivirus optimization strategies are proposed: (i) risk-based scan frequency, (ii) signature scan repetition avoidance, and (iii) selective signature scanning. Evaluations show that these strategies can reduce antivirus energy consumption by up to 56% without degrading security. Additional recommendations include aligning scans with user activity to avoid energy-intensive idle-time operations.

Shafique et al. [61] propose a cross-layer framework for developing energy-efficient and secure Edge AI systems that addresses performance, energy efficiency, reliability, and security challenges for deep neural networks (DNNs) and spiking neural networks (SNNs) on resource-constrained devices. The framework integrates hardware/software optimizations, including pruning, quantization, approximate computing, and neural architecture search, where RL identifies optimal architectures. Reliability is enhanced through fault-aware training, range restriction, and aging mitigation, while security is fortified via data encryption, side-channel attack defenses, and hardware obfuscation. The paper reports individual technique gains, including up to 88% energy reduction and significant latency improvements. For SNNs, methods such as SparkXD and ReSpawn achieve up to a 70% improvement in accuracy under memory faults. AI methodologies, including DL, RL, and neuromorphic computing, are core to the framework, which holistically combines performance, robustness, and sustainability in edge AI deployment.

Green IoT studies converge on the principle that sustainability must be designed from the hardware layer upward. They emphasize minimizing redundant computation, optimizing routing, and embedding AI models that adapt dynamically to energy availability. This cluster exemplifies a shift toward cross-layer co-design, integrating hardware, AI, and network management to create truly sustainable smart-city ecosystems.

3.5. Optimized Energy Consumption in Networks and Protocols

Khattak [62] examines cybersecurity vulnerabilities in Cooperative Driving Automation (CDA) in smart cities, with a particular focus on Vehicle-to-Infrastructure (V2I) communications. The authors developed a simulation framework that integrates optimal control, Cooperative Adaptive Cruise Control (CACC), and ML to evaluate the impact of three cyberattacks: fake basic safety messages, replay attacks, and DoS attacks. Using VISSIM and Autonomie for traffic and energy modeling, they found that Fake BSMs increased acceleration-deceleration volatility by 35-37%,

Table 1: Summary of State-of-the-Art AI-Cybersecurity Approaches

Study	Advantages	Disadvantages	Energy/Sustainability
[35]	Selective client participation improves energy efficiency	Needs tuning of entropy-energy weights	96.7% energy reduction vs FedAvg
[36]	Identifies key sustainability challenges in FL	No mitigation strategies proposed	FL emits up to 100x CO_2 vs centralized
[37]	Reduces communication drastically	Limited to simple pattern recognition	Doubles IoT network lifetime
[38]	Improves security and reduces rounds	Encryption overhead not quantified	Reduced communication energy
[39]	Supports client personalization	Increased model complexity	32.3% energy efficiency gain
[40]	Dynamic, secure, low-latency framework	Needs DRL tuning and blockchain overhead	47.5% local execution with 1MHz bandwidth
[41]	Enhanced security and throughput in WSNs	Increased protocol complexity	52.73% energy reduction
[42]	Trust-based energy-aware clustering and sensing	Blockchain integration adds latency	Efficient energy trading, reduced false detection
[43]	Reduces data redundancy and ensures data integrity	Linear gas cost growth	35% energy savings
[44]	Strong IDS integration with SDN and blockchain	High initial setup complexity	50% energy savings
[45]	Effective against node capture and replay attacks	Slight throughput reduction	60% less data manipulation probability
[46]	Certificate-based dynamic authentication	Blockchain adds 21.2% operation time	39.07% energy reduction
[47]	Secure and efficient CH selection	Increased system overhead	Reduced power consumption
[23]	Real-time adaptive IDS with DRL	Requires retraining for unseen attacks	Energy-efficient with high detection accuracy
[48]	Fast, low-energy botnet detection	Memory usage is moderately high	~3.5J per detection, minimal carbon emission
[22]	Dynamic security policy selection	Policy tuning needed	Low energy and rapid IDS response
[49]	Adaptive DDoS mitigation in 5G	Simulation-based validation	Lower energy during normal operations
[50]	Real-time energy anomaly detection	Focuses only on overconsumption	Low overhead, real-time optimization
[51]	Robust zero-day attack detection	Relies on synthetic attack data	Low false alarm, efficient online deployment
[52]	High-accuracy CNN anomaly detection	Slight overfitting observed	93.64% detection accuracy
[53]	Dual-optimization for load management	Requires high-quality input data	46.9% carbon, 43.9% energy reduction
[54]	Ultra-low latency with DVNs	Specialized hardware needed	Massive energy and cost savings
[55]	Adaptive threat mitigation	Requires game-theoretic modeling	Energy-aware dynamic defenses
[56]	Hardware-accelerated ML IDS	Reduced model flexibility	Up to 93% energy savings
[57]	Optimized classifier via swarm-based boosting	Moderate model complexity	High accuracy, reduced resource strain
[61]	Comprehensive energy-robust AI design	Complex multi-technique framework	Up to 88% energy savings
[58]	Secure edge routing with low energy cost	Security scalability may vary	21% energy reduction
[59]	TinyML for sustainable IoT edge	No new AI method introduced	Reduced carbon and bandwidth use
[60]	Energy-optimization strategies for antivirus	Requires tailored policy per AV tool	56% AV energy savings
[62]	Quantifies attack-induced energy waste	Focuses on simulation only	Emissions increase by 15–23% due to attacks
[63]	Safety-guaranteed controller switching	Relies on accurate model abstractions	Lowered energy cost while ensuring safety
[64]	Low-memory CNN training	Specialized setup needed	Up to 7x carbon reduction
[65]	Compares ML models for energy-aware IDS	Deep models less practical	DT most efficient in SCADA/Edge
[66]	High F-measure detection of IoT devices	Dataset may limit generalization	Energy optimization via idle detection
[67]	Clustering and ML-based IDS	Lower accuracy in real-time tests	Improved energy and throughput

while replay and DoS attacks increased it by 26-34%. Consequently, emissions rose by 15-23% and fuel consumption by 3-13%. The study also introduced the LSTM neural network for real-time attack detection, achieving 97%-98% accuracy. The results emphasize that cyberattacks degrade flow stability, environmental sustainability, and safety, underscoring the need for resilient monitoring algorithms in future smart cities.

Wang et al. [63] propose an adaptive control framework for Learning-Enabled Cyber-Physical Systems (LE-CPSs) that guarantees safety while optimizing energy efficiency.

The method integrates formal methods and ML by approximating each controller, including neural networks, using Bernstein polynomials to compute robust invariant sets defining safe operational spaces. The union of these sets constitutes a joint safe configuration space. A Double Deep Q-Network (Double DQN) is developed to learn an optimal switching strategy between controllers, minimizing control energy consumption while enforcing safety via a penalty mechanism for unsafe choices. The framework is validated on a Van der Pol oscillator and an Adaptive Cruise Control (ACC) system. For the oscillator, the approach achieved a 100% safety rate with an average energy cost of 127.8, outperforming baselines (130.1 to 383.8). In the ACC system, energy consumption was reduced to 835.7 compared to 854.8-1085.5 for other methods, while maintaining complete safety. Additionally, the technique demonstrated robustness against disturbances up to twice the design assumption, sustaining high safety rates.

Anandharaju et al. [64] propose HotConv, a low-carbon footprint training framework for 1D CNNs that features temporal max-pooling, a technique prevalent in applications such as malware detection and genomics. HotConv exploits the sparsity of max-pooling layers by caching only the maximum activations and corresponding gradients, substantially reducing GPU memory requirements without degrading model performance. A Partitioned Data Loading (PDL) strategy is also introduced to optimize data handling and minimize training time. Experiments using the BODMAS and VirusTotal malware datasets on the MalConv architecture showed that HotConv reduces GPU memory usage to $\frac{1}{2}$ and $\text{HotConv}_{\text{Eco}}$ to $\frac{1}{22}$ of the memory required by the optimized MalConv2, cutting carbon emissions up to $\frac{1}{3}$ and $\frac{1}{7}$, respectively. Specifically, $\text{HotConv}_{\text{Eco}}$ achieved a total carbon footprint of 0.45 $kgCO_2e$ over 100 epochs, compared to 3.25 $kgCO_2e$ for MalConv2, while preserving a comparable Precision-Recall Curve Area Under the Curve (PRC AUC). Moreover, under a 512MB GPU constraint, $\text{HotConv}_{\text{Eco}}$ processed complete sequences without truncation, enhancing PRC AUC from 0.83 (MalConv2) to 0.89 on BODMAS and from 0.87 to 0.94 on VirusTotal.

Chijioke Ahakonye et al. [65] present Eco-Secure SCADA, a framework that evaluates energy-efficient ML models for intrusion detection in SCADA, IIoT, and Edge IoT networks, aiming to enhance both cybersecurity and sustainability. The study benchmarks models such as DT, RF, LightGBM, KNN, neural networks (NN), and CNN-LSTM using datasets including Edge-IIoTset, WUSTL-IIoT-2021, and ICS-SCADA. DT emerged as the most energy-efficient, achieving efficiency scores of 13.03 (SCADA), 19.05 (IIoT), and 83.85 (Edge IoT), while maintaining high accuracy (up to 90-99%). In contrast, CNN-LSTM and NN achieved high accuracy but incurred excessive training and inference times, limiting their practicality in resource-constrained environments. DT and LightGBM exhibited minimal energy cost with fast inference, making them ideal for real-time, low-power applications.

Algarni et al. [66] propose a ML-based framework to enhance security and energy efficiency in Smart Energy Management Systems (SEMSs) by detecting IoT devices within a network. Utilizing a dataset containing 13,138 records across five device types with 18 extracted features, the paper employed NB and DT. The DT model achieved a superior weighted F-measure of 99% compared to 91% for NB. The system was validated via penetration testing, vulnerability assessments, and simulated DDoS attacks, confirming its robustness against cybersecurity threats. Feature importance analysis improved model transparency and decision-making. Energy efficiency was addressed by identifying idle and compromised devices, enabling optimizations such as sleep modes and advanced power management.

Gulganwa et al. [67] present the EES-WCA algorithm, an Energy-Efficient and Secure Weighted Clustering Algorithm for WSNs that integrates ML-based IDS. The system combines the EE-WCA clustering method for network scalability and energy optimization with a centralized IDS at the base station. The IDS utilizes Support Vector Machine (SVM) and Multi-Layer Perceptron (MLP) classifiers to detect six WSN attacks: black hole, wormhole, selective forwarding, HELLO flood, jamming, and exhaustion. A dataset generated from NS2.35 simulations includes features such as total sent packets, forwarded packets, RTT, packet drop rate, energy variation, and buffer length. The IDS achieved up to 90% detection accuracy in validation tests and approximately 77% in real-time scenarios, with the MLP demonstrating superior performance compared to SVM. Network evaluations revealed improvements in Packet Delivery Ratio, throughput, end-to-end delay, and energy consumption compared to the baseline EE-WCA.

Studies in this category bridge AI-driven optimization and sustainable operations at the protocol and network level. They demonstrate that security and sustainability can be co-achieved through energy-aware scheduling, adaptive control, and low-carbon model training. The emerging trend favors hybrid frameworks combining ML with control theory to maintain both safety and ecological efficiency.

Table 1 presents a summary of the reviewed state-of-the-art studies, emphasizing the advantages, disadvantages, and reported energy and sustainability gains of each work. The selected papers reflect diverse AI-driven approaches applied to cybersecurity with clear sustainability goals, ranging from energy-efficient FL to blockchain-secured IoT networks and lightweight IDS. This comparative overview provides a holistic understanding of the current strengths and limitations in designing secure, low-power, and environmentally sustainable digital infrastructures.

Recent works, such as [40, 57, 62], further enrich this landscape. The study in [57] applies DL to malware detection in IoT, offering insights into scalable, adaptive model architectures for cyber defense. Work on cooperative and resilient networked systems [62] demonstrates how distributed intelligence enhances robustness and sustainability in the face of cyber-physical stress. Meanwhile, [40] introduces adaptive, energy-efficient routing and resource optimization

Additionally, the reviewed studies collectively demonstrate a growing convergence between AI-driven cybersecurity and sustainability objectives. Applying the defined sustainability metrics across clusters revealed consistent trade-offs between security accuracy, communication efficiency, and environmental impact. ML and blockchain remain the dominant paradigms; however, emerging fields such as neuromorphic computing, lightweight AI, and LLMs present untapped potential for achieving greener, adaptive, and scalable security. This synthesis lays the groundwork for the conceptual framework and research agenda presented in subsequent sections.

3.6. Comparative Sustainability Analysis of AI Methods

This section compares the reviewed AI-based cybersecurity approaches across standard sustainability metrics, including energy efficiency, carbon footprint, and resource utilization. The goal is to highlight trade-offs and identify methods that balance strong security performance with sustainable operation.

Figure 6 demonstrates that DRL achieves the highest balance across all three sustainability dimensions, particularly excelling in energy efficiency and resource optimization. FL follows closely, though it incurs a moderate carbon footprint due to its decentralized nature and communication overhead. Blockchain exhibits significant limitations, scoring low in all sustainability aspects due to its inherently energy-intensive cryptographic processes. AEs make moderate contributions to energy efficiency and resource optimization, but they remain limited in terms of awareness of their carbon footprint. Lastly, LLMs perform worst in sustainability metrics, reflecting their substantial energy demands and limited resource optimization in current applications. This comparative insight underscores the pressing need to optimize LLMs for sustainability and to adapt blockchain methods for greener cybersecurity solutions.

Figure 7 illustrates the distribution of key metrics, including energy consumption, carbon emission, CPU usage, sustainability, and privacy preservation. These metrics were selected based on recurring evaluation criteria identified in the reviewed literature (see Section 2) and their relevance to both cybersecurity performance and sustainability objectives, thereby addressing a significant gap: prior studies often emphasized accuracy without considering environmental and resource-related impacts. FL exhibits strong energy efficiency and excels in privacy preservation. At the same time, DRL offers a balanced trade-off across all metrics, albeit with moderate CPU usage and carbon emissions. Blockchain-based techniques offer strong privacy protection but incur high carbon emissions and significant CPU usage, limiting their applicability in resource-constrained environments. AEs demonstrate consistent yet moderate performance across all sustainability aspects, indicating their utility in general-purpose scenarios. LLMs, despite their advanced computational capabilities, exhibit the highest CPU usage and carbon emissions, rendering them less suitable for sustainability-focused applications.

Figure 8 maps the energy efficiency of AI methods against their carbon awareness, with visual encodings representing resource intensity and deployment costs. This perspective underscores that methods, e.g., spiking neural networks and lightweight ML algorithms, offer superior sustainability profiles, whereas blockchain and LLMs present significant environmental and computational overheads. Furthermore, these analyses inform AI selection for cybersecurity by highlighting the inherent trade-offs among sustainability, computational efficiency, and security performance.

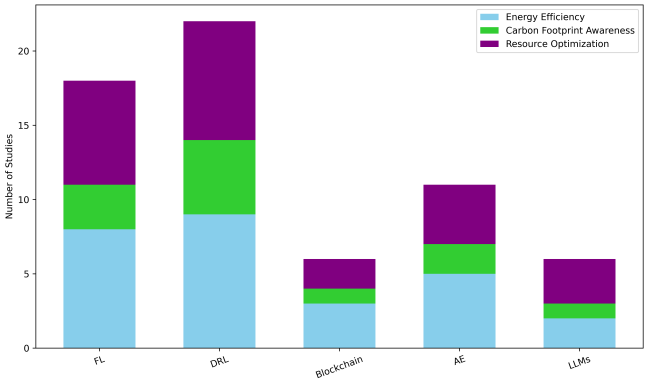


Figure 6: A comparative analysis of energy efficiency, carbon footprint awareness, and resource optimization across various AI approaches, e.g., FL, DRL, blockchain, AE, and LLMs.

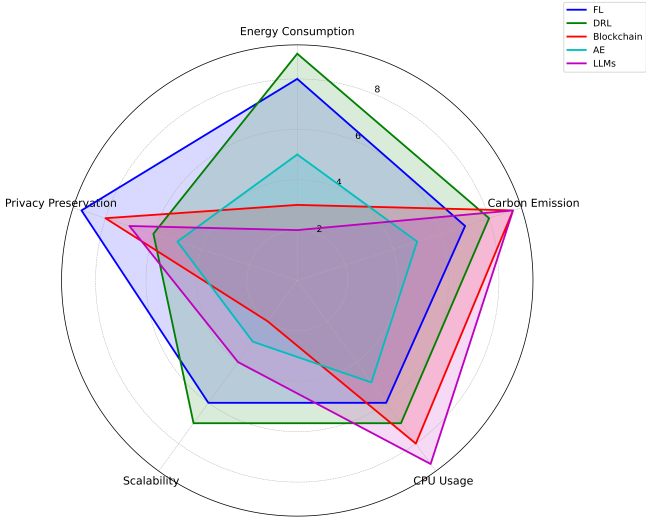


Figure 7: The comparison highlights AI methods across key sustainability and performance metrics, including energy consumption, carbon emissions, CPU usage, scalability, and privacy preservation.

4. Sustainability, Security Trade-Off Analysis

Although Section 3 categorizes existing AI-based cybersecurity approaches according to techniques and application domains, a deeper synthesis of the reviewed literature reveals fundamental trade-offs between cybersecurity effectiveness

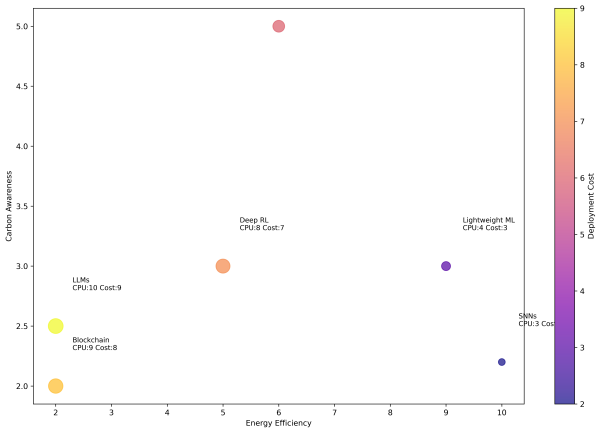


Figure 8: The relationship between energy efficiency and carbon awareness in various AI methods is represented, with resource and CPU usage scaled proportionally.

and sustainability objectives that are rarely explicitly articulated. These trade-offs play a decisive role in determining whether AI-driven security mechanisms can be realistically deployed within resource-constrained, environmentally regulated infrastructures. Rather than reinforcing each other by default, security performance, energy efficiency, privacy preservation, and environmental impact frequently exist in tension, creating design contradictions that current research only partially addresses.

A central trade-off emerges between detection performance and energy consumption. DL and transformer-based intrusion detection systems consistently demonstrate superior detection accuracy and robustness against complex and evolving attack patterns. However, this performance advantage comes at the cost of high computational complexity, repeated training cycles, and energy-intensive inference, resulting in increased power consumption and higher carbon emissions. Studies based on DRL further amplify this effect, as adaptive policy learning requires prolonged training and frequent updates. In contrast, lightweight ML models, including decision trees, shallow neural networks, and hybrid approaches, exhibit lower detection accuracy in highly adversarial environments; however, they offer substantially reduced energy consumption, faster inference, and improved suitability for long-term deployment on edge and IoT devices. This contrast illustrates that optimizing solely for detection accuracy often produces solutions that are misaligned with sustainability requirements in real-world infrastructures.

A second critical tension exists between privacy preservation and environmental impact. Privacy-preserving mechanisms such as FL, secure aggregation, and encrypted communication reduce data exposure and regulatory risk by keeping sensitive information decentralized. Nevertheless, these mechanisms introduce additional computation and communication overhead, which can significantly increase energy consumption and carbon footprint when not carefully optimized. Empirical evidence suggests that FL may result in higher emissions than centralized training when communication rounds are frequent and client participation is inefficient. This contradiction highlights that privacy enhancement does not inherently imply sustainability improvement, underscoring the need for joint optimization strategies that explicitly consider both objectives.

Blockchain-based cybersecurity solutions further expose the trade-off between decentralized trust and computational overhead. Blockchain enhances integrity, transparency, and non-repudiation in distributed systems, yet consensus mechanisms and cryptographic operations impose substantial energy and storage costs. Even energy-aware blockchain variants reduce consumption only marginally when compared to lightweight security protocols. As a result, blockchain-based security architectures remain difficult to justify in sustainability-sensitive deployments unless applied selectively, for example, for audit logging and access control rather than continuous threat detection. The limited availability of end-to-end environmental cost assessments for blockchain-enabled security systems further obscures informed decision-making.

Adaptability introduces another layer of tension between resilience and sustainability. Adaptive security models, based on online learning and RL, dynamically adjust defense strategies in response to evolving threats and changing resource availability, thereby enhancing robustness and responsiveness. However, frequent model updates, retraining, and policy adjustments increase computational load, energy consumption, and lifecycle emissions. Static models, by comparison, offer predictable resource consumption but lack responsiveness to novel attack patterns. The literature suggests that adaptability must be constrained through energy budgets, retraining thresholds, and lifecycle-aware optimization to prevent uncontrolled sustainability costs.

Hardware-aware optimization highlights a trade-off between energy efficiency and deployment generality. Specialized accelerators, including FPGAs and neuromorphic processors, significantly reduce energy consumption per impact when compared to general-purpose CPUs and GPUs. However, these gains come at the expense of portability, scalability, and system complexity, limiting applicability across heterogeneous infrastructures. Software-only solutions remain more flexible but incur higher energy costs. This contrast underscores the importance of cross-layer co-design, aligning model architecture, hardware capabilities, and deployment context, a practice that remains underexplored in existing cybersecurity research.

5. Research Agenda and Conceptual Framework for Sustainable Cybersecurity

Despite growing interest in energy-efficient cybersecurity, existing research remains fragmented across domains such as FL, blockchain, and ML-based IDS. Although many works explore AI-based methods (e.g., ML) and blockchain, these are often presented in isolation, without a unified structure. To address this fragmentation, a unified

conceptual framework is proposed, integrating three interconnected layers: (i) a cybersecurity resilience layer, encompassing threat detection and privacy-preserving mechanisms,i.e., secure FL; (ii) a sustainability monitoring layer, focused on carbon awareness, energy usage tracking, and hardware-aware training; and (iii) an AI optimization layer, leveraging model compression, pruning, and low-power architectures. The inclusion of specific techniques,i.e., FL and blockchain, in this framework is illustrative rather than prescriptive; other methods (e.g., secure multi-party computation, lightweight encryption, and centralized ML pipelines) may be substituted depending on the application requirements, threat model, and resource constraints. This framework is explicitly aligned with the core sustainability metrics defined in Section 2, namely, energy consumption (J/inference), carbon emissions (CO_2/epoch), computational efficiency (ops/Watt), and privacy preservation, ensuring consistent evaluation across reviewed approaches.

The framework is aligned with recent contributions by Jamshidi et al. [23], Qiu et al. [36], and Alsulami et al. [57]. Specifically, Jamshidi et al. propose self-adaptive security mechanisms for dynamic IoT networks, emphasizing automated model selection and retraining, which is reflected in our AI optimization layer. Qiu et al. introduce the first carbon-aware AI security benchmarking, directly informing the sustainability monitoring layer through energy and emissions tracking. Alsulami et al. present a privacy-preserving FL architecture for edge computing, elements of which are incorporated in the cybersecurity resilience layer of our framework. By integrating these complementary ideas, the proposed framework enables a holistic evaluation of security models that jointly consider performance, privacy, and environmental impact. This integrated perspective is particularly advantageous in IoT systems, where devices often operate under strict energy and computational constraints while handling sensitive data, necessitating solutions that are simultaneously secure, sustainable, and resource-aware. For example, in a smart grid deployment, the cybersecurity resilience layer could employ FL-based anomaly detection to secure edge nodes. At the same time, blockchain and lightweight ledger systems log critical events for auditability. The sustainability monitoring layer would continuously track per-node energy usage and CO_2 footprint. The AI optimization layer could adaptively prune or quantize models during low-power conditions, ensuring uninterrupted security while staying within sustainability thresholds. However, while the framework conceptually connects AI-enabled security and sustainability, it previously lacked concrete operational metrics. To strengthen its practical applicability, measurable indicators have been incorporated, including carbon footprint (kg CO_2e/epoch), power usage effectiveness (PUE), and energy consumption per inference (J/inference). These quantifiable parameters enable objective evaluation and benchmarking of "green" cybersecurity solutions across heterogeneous platforms, transforming the high-level framework into an actionable tool for empirical validation and cross-domain comparison. A summary of the framework's core components and associated techniques is provided in Table 2.

5.1. Mapping Sustainability Objectives in AI-Based Cybersecurity

To address RQ2, this section investigates the extent to which AI-based cybersecurity approaches incorporate sustainability objectives, including energy efficiency, resource constraints, and environmental impact. The analysis examines how these factors are integrated into the design, deployment, and evaluation of security systems.

Figure 9 illustrates a multi-dimensional sustainability architecture for AI-driven cybersecurity systems. At the center of the framework lies the *AI-Driven Cybersecurity Engine*, which comprises core technical mechanisms such as intrusion detection systems, LLM-based threat analysis, and machine learning/deep learning models, evaluated using performance metrics (e.g., accuracy and F1-score). This central component reflects the conventional objective of maximizing security effectiveness. Surrounding the core are four interdependent sustainability dimensions. The *Computational Sustainability* dimension captures operational resource requirements, including energy consumption, GPU utilization, model size, training versus inference cost, and deployment strategies (edge versus cloud). These factors directly influence the system's resource efficiency. The *Environmental Sustainability* dimension reflects the broader ecological footprint of AI-based cybersecurity systems, incorporating carbon emissions, carbon intensity factors, lifecycle assessment (LCA), cooling infrastructure overhead, hardware manufacturing impact, and electronic waste generation. This layer highlights that sustainability extends beyond immediate power usage to full lifecycle considerations. The *Economic Viability* dimension emphasizes operational cost, energy expenditure, infrastructure investment, and cost–performance trade-offs, which collectively determine long-term feasibility and scalability of secure AI deployments. Moreover, the *Socio-Technical & Policy Alignment* dimension integrates governance-related considerations, including SDGs alignment, responsible AI principles, transparency, regulatory compliance, and ethical deployment.

5.1.1. Evaluation of Energy Efficiency, Resource Constraints, and Environmental Impact

Table 3 presents a mapping of sustainability objectives, namely energy efficiency, resource constraints, and environmental impact, across state-of-the-art AI-based cybersecurity studies. A dominant observation is the widespread emphasis on energy efficiency, with over 85% of the reviewed studies explicitly targeting reductions in energy consumption through algorithmic optimizations, hardware-aware designs, and protocol-level enhancements. This reflects a growing consensus on the need to address the energy demands of AI-driven cybersecurity, particularly in resource-constrained environments such as IoT ecosystems, edge computing platforms, and wireless sensor networks.

Table 2
Proposed Conceptual Framework for Sustainable Cybersecurity

Framework Layer	Main Responsibilities and Techniques (Examples)
Cybersecurity Resilience Layer	• ML-based IDS for anomaly detection and classification • Blockchain for secure logging and authentication (e.g., tamper-proof audit trails), optional, depending on application needs • FL for privacy-preserving model updates across distributed nodes • Real-time threat response and anomaly detection under constrained resources • Alternative methods: secure multi-party computation, lightweight encryption, centralized ML pipelines when appropriate
Sustainability Monitoring Layer	• Measurement of energy usage and hardware load, expressed in defined core metrics (e.g., J/inference, ops/Watt) • Carbon emission modeling and reporting (e.g., CO_2 per training round, CO_2/epoch) • Green benchmarking across devices and environments for fair cross-study comparisons • Integration of sustainability thresholds into operational decision-making
AI Optimization Layer	• Model pruning, quantization, and knowledge distillation to reduce size and energy usage • Hardware-aware training, adaptive inference scheduling, and dynamic model selection • Low-power edge model design (TinyML, efficient CNNs, spiking neural networks) • Multi-objective optimization balancing accuracy, latency, and sustainability metrics

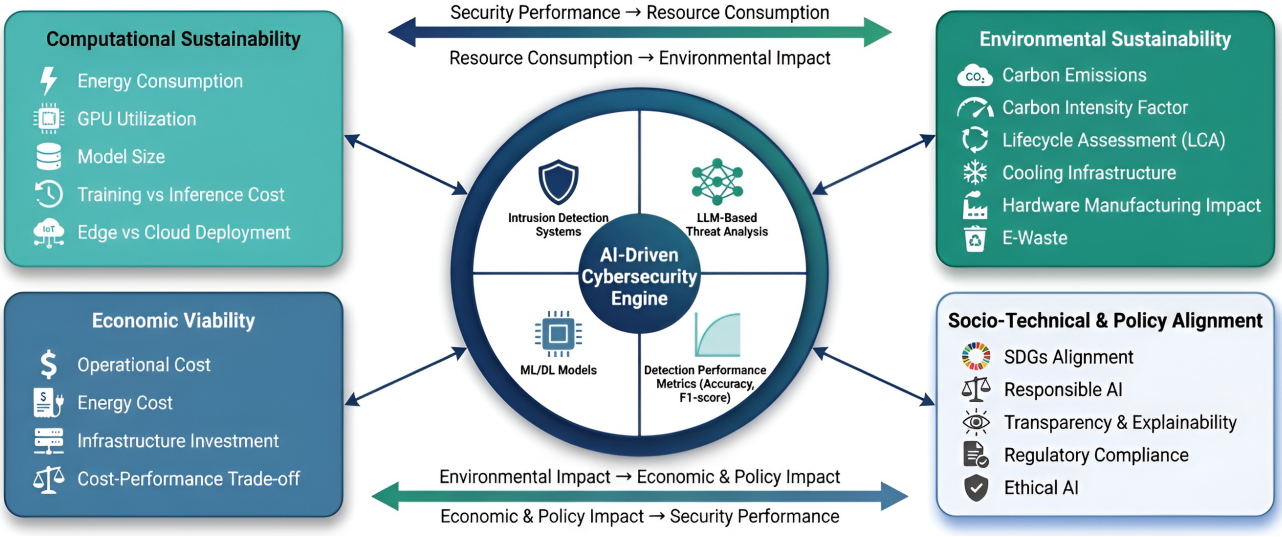


Figure 9: Multi-dimensional sustainability framework for AI-driven cybersecurity, linking security performance with computational resource consumption, environmental lifecycle impact, economic viability, and socio-technical & policy alignment.

Despite these advances, a holistic consideration of resource constraints, including bandwidth, memory, and processing power, remains relatively limited. Important exceptions include studies leveraging FL (e.g., [35], [37]) and blockchain-based security frameworks ([42], [43]), which inherently account for distributed resources. Nevertheless, resource-aware security models remain scarce, a gap that is increasingly critical as AI models become more complex and deployment scenarios demand greater efficiency on low-power devices. Furthermore, environmental impact metrics, e.g, carbon footprint and CO_2 emissions, are addressed in only a minority of studies (e.g., [36], [48], [53], [64]). This suggests that while there is an emerging interest in sustainability-aware security, systematic assessments of environmental consequences remain underdeveloped. Most current works rely on indirect measures, such as energy

Table 3
Detailed Sustainability Objectives in AI-based Cybersecurity Approaches

Study	Energy Efficiency	Resource Constraints	Environmental Impact	Metrics Reported
[35]	Yes (Energy consumption)	Yes	No	Energy consumption (Wh)
[36]	Yes	Yes	Yes (CO2 emissions)	CO2 emissions (g)
[37]	Yes	Yes	No	Network lifetime
[38]	Yes (Comm. energy cost)	Yes	No	Communication energy cost
[39]	Yes (Energy efficiency %)	Yes	No	Energy efficiency percentage
[40]	Yes	Yes	No	Local execution ratio
[41]	Yes (Energy reduction %)	Yes	No	Energy reduction %
[42]	Yes (Sensing gain, energy)	Yes	No	Sensing gain, energy
[43]	Yes (Energy & time delay)	Yes	No	Energy & time delay
[44]	Yes (Energy, CPU usage)	Yes	No	Energy, CPU usage
[45]	No	Yes	No	Data manipulation probability
[46]	Yes (Energy consumption)	Yes	No	Energy consumption
[47]	Yes	Yes	No	Power consumption
[23] [48] [22]	Yes (Energy, carbon emissions)	Yes	Yes	Energy, carbon emissions
[49]	Yes	Yes	No	Throughput, detection rate
[50]	Yes	Yes	No	MAE, RMSE
[51]	Yes	Yes	No	Detection rate, False Alarm rate
[52]	Yes	Yes	No	Accuracy
[53]	Yes (Energy, carbon, resource use)	Yes	Yes	Energy, carbon, resource utilization
[54]	No	No	No	Link utilization, latency
[55]	Yes	Yes	No	Conceptual
[56]	Yes (Energy profiling)	Yes	No	Energy profiling
[57]	Yes (Accuracy, energy)	Yes	No	Accuracy, energy
[61]	Yes (Energy, latency)	Yes	Yes	Energy, latency
[58]	Yes	Yes	Yes	Throughput, delay
[59]	Yes	Yes	Yes	Carbon footprint
[60]	Yes (Energy savings in AV)	Yes	Yes	Energy savings in AV
[62]	No	Yes	Yes	Emissions, fuel consumption
[63]	Yes (Energy cost metric)	Yes	Yes	Energy cost metric
[64]	Yes	Yes	Yes	Carbon emissions (kg)
[65]	Yes (Energy scores)	Yes	No	Energy scores
[66]	Yes (F-measure, energy)	Yes	No	F-measure, energy
[67]	Yes (Energy consumption)	Yes	No	Energy consumption

consumption, rather than adopting direct carbon accounting and environmental sustainability metrics. Additionally, the metrics employed to evaluate these systems are highly heterogeneous, spanning from energy consumption (e.g., in Watt-hours) and network longevity to carbon emissions and throughput performance. This variability highlights the lack of standardized evaluation frameworks that can simultaneously capture both security efficacy and sustainability performance. For instance, Jamshidi et al. [48] provide a more holistic sustainability assessment by reporting both energy usage and carbon emissions in edge security deployments. Yet, such integrative evaluations are the exception rather than the norm.

5.2. Taxonomy of AI Techniques, Security Domains, and Sustainability Focus

Table 4 presents a structured taxonomy that categorizes the reviewed studies based on three dimensions: the AI techniques employed, their specific cybersecurity application domains, and the primary focus on sustainability. The analysis reveals that FL and DRL are increasingly favored for resource-constrained environments, e.g, IoT and edge devices, as seen in works by Savoia et al. [35] and Jamshidi et al. [48]. These methods are particularly well-suited for decentralized security mechanisms, ensuring data privacy while optimizing energy consumption. Moreover, DL approaches, including CNNs, AEs, and LSTM models, dominate in applications, e.g, anomaly detection, malware classification, and smart grid IDS. Studies, e.g., those by Durai et al. [64] and Takiddin et al. [51] leverage these models to achieve both security and computational efficiency, with explicit reductions in carbon footprint reported in some cases. Additionally, blockchain-integrated models emerge as robust solutions for securing IoT ecosystems, enhancing data integrity and transparency while managing energy costs, as exemplified by works such as Islam et al. [47] and Goyat et al. [46].

The taxonomy also highlights the growing focus on diverse cybersecurity domains, ranging from IDS to smart-city transportation and SCADA systems. Yet, the sustainability focus varies widely: some studies prioritize energy savings (e.g., Gurram et al. [41]), while others focus on carbon footprint (e.g., Saxena et al. [53]), and fewer still integrate comprehensive resource-efficiency metrics. Additionally, while methodological innovation is evident, a gap remains in fully integrating multidimensional sustainability metrics, including environmental impact, resource constraints, and lifecycle energy costs, alongside security performance.

5.3. Integrating Cybersecurity and Sustainability: Methods, Metrics, and Applications

Table 5 synthesizes the diversity of AI-based cybersecurity research, illustrating how different methodologies align with sustainability metrics across various application domains. A key observation is the prominence of DRL and FL approaches in achieving both cybersecurity robustness

Table 4
Classification of AI-based Cybersecurity Studies by Technique, Domain, and Sustainability Focus

Study	AI Technique	Cybersecurity Domain	Primary Sustainability Focus
Savoia et al. [35]	FL	Edge Security	Energy-aware client selection
Qiu et al. [36]	FL, Analytical Modeling	General Security Systems	Carbon footprint estimation
Jamshidi et al. [48]	DRL	IoT Edge IDS	Energy and carbon efficiency
Saxena et al. [53]	Neural Networks and Heuristics	Cloud Data Center Security	Energy and carbon reduction
Durai et al. [64]	CNNs with Memory Optimization	Malware Detection	Low-carbon AI training
Brudni et al. [60]	ML Regression Models	Antivirus Optimization	Energy optimization in AV
Fraga-Lamas et al. [59]	Edge AI, TinyML	Green IoT Security	Carbon footprint in edge AI
Wang et al. [63]	Double Deep Q-Network	CPS Safety	Energy-optimal control with safety
Khattak [62]	LSTM and Smart Transportation Security	Emissions	Safety under attacks
Alsulami et al. [57]	Weight-normalized AdaBoost	IoT Threat Detection	Energy reduction via optimized classifiers
Viegas et al. [56]	DT, HW Acceleration	IDS for Embedded Systems	Hardware energy profiling
Gurram et al. [41]	AE and Metaheuristics	Secure WSN Routing	Reduced energy consumption
Maiwada et al. [49]	Statistical Analysis	5G IDS	Energy-aware DDoS mitigation
Ahakonye et al. [65]	DT, LightGBM	SCADA/IIoT IDS	Low-energy inference
Takiddin et al. [51]	AE with Attention	Smart Grid Theft Detection	Energy-efficient detection
Halabi et al. [55]	Conceptual RL	Green IoT Security	Adaptive, low-cost security
Shafique et al. [61]	DNN, SNN, NAS	Edge AI Security	Cross-layer energy savings
Islam et al. [47]	Blockchain, NFV	IoT Secure Routing	Energy-efficient network
Goyat et al. [46]	Blockchain + ML	Green IoT Privacy	Energy-efficient data handling

and energy efficiency. For instance, Jamshidi et al. [48] and Savoia et al. [35] effectively employ DRL and FL to optimize IDS and client selection while significantly reducing energy consumption in edge and distributed environments. Such methods are particularly well-suited for decentralized, resource-constrained infrastructures, e.g, IoT and edge computing. Moreover, blockchain-based security solutions also emerge as significant contributors to sustainability-conscious security, as seen in Aldawsari et al. [42] and Gurram et al. [41]. These approaches not only enhance security through decentralization and trust mechanisms but also incorporate energy-aware clustering and resource optimization to improve network sustainability and resilience. The evaluation metrics reported across studies vary, including energy consumption in watt-hours, carbon emissions, throughput, latency, and resource utilization. However, there is an apparent absence of standardized sustainability key performance indicators. For example, Qiu et al. [36] provide an explicit carbon footprint analysis of FL, whereas other studies focus on relative energy savings or operational performance without direct environmental metrics. This heterogeneity in reporting impedes holistic assessment and cross-study comparisons. In terms of application domains, most research efforts focus on IoT systems, smart grids, and edge computing environments, where energy constraints and environmental impacts are particularly critical. Yet, domains such as smart transportation and healthcare systems, which also demand secure and sustainable solutions, remain comparatively underexplored in current literature. Furthermore, these insights emphasize the need for unified frameworks that integrate lifecycle carbon assessments, sector-specific sustainability benchmarks, and adaptive AI models that co-optimize cybersecurity efficacy and environmental performance.

5.4. Comparative Analysis of AI Methods in Sustainable Cybersecurity

The integration of Artificial Intelligence into cybersecurity has ushered in a new era of intelligent defense mechanisms, yet the degree to which these approaches embody sustainability varies widely across techniques and application domains. By aligning AI methodologies with explicit sustainability metrics and sector-specific applications (Table 5) and classifying AI techniques by cybersecurity domain and focus (Table 4), a clearer understanding emerges of how different technologies contribute to both protection and environmental responsibility. The analysis reveals that DRL and FL dominate in resource-constrained IoT and edge environments, where they effectively balance security robustness with energy efficiency. Conversely, blockchain-based systems have proven instrumental in safeguarding data integrity and ensuring decentralized trust in smart cities, though their high computational and energy demands continue to raise sustainability concerns. Despite these achievements, the generalization of sustainability-aware cybersecurity strategies remains elusive. Critical domains such as healthcare, smart transportation, and water infrastructure still lack integrated approaches that jointly optimize security performance and ecological impact.

Each AI paradigm brings its own promise and paradox. FL empowers distributed intelligence by enabling local model training without centralized data sharing, thereby preserving privacy and reducing the carbon footprint associated with large-scale data transfer. Yet, the process introduces synchronization and aggregation costs that can offset energy gains if not carefully optimized. Techniques such as energy-aware scheduling and selective client participation are essential to realize FL's full sustainability potential. Similarly, DRL excels in dynamic, threat-rich environments where adaptive decision-making is critical. Its ability to learn optimal defense policies in real time represents a milestone in proactive cybersecurity. However, DRL's extensive training cycles and computational intensity remain a sustainability

Table 5
AI-based Cybersecurity Approaches: Methodologies, Sustainability Metrics, and Application Domains

Study	Methodology	Sustainability Metrics	Application Domain
Jamshidi et al. [48]	DRL	Energy consumption (j), Carbon emissions (kg)	IoT Edge Security
Savoia et al. [35]	FL	Energy consumption (Wh), Client energy status	Edge Computing, Distributed Learning
Qiu et al. [36]	FL Carbon Footprint Model	CO_2 emissions (g), Energy profiling	FL Systems
Aldawsari et al. [42]	Blockchain with Cognitive Radio	Energy-aware clustering, Spectrum efficiency	Smart Cities, WSNs
Gurram et al. [41]	Meta-Heuristic Routing with Blockchain	Network throughput, Energy consumption reduction	IoT-based WSNs
Ahmed et al. [43]	Blockchain and Fuzzy Clustering	Energy savings (%), Data aggregation accuracy	IoT Data Aggregation
Saxena et al. [53]	Neural Networks with Meta-heuristics	Carbon footprint reduction, Power Usage Effectiveness	Cloud Data Centers
Wang et al. [63]	Adaptive Control with DRL	Energy cost optimization, Safety guarantees	Learning-Enabled CPS (e.g., autonomous systems)
Fraga-Lamas et al. [59]	Green IoT and Edge AI	Carbon footprint, Energy consumption reduction	Smart Circular Economy, Industry 5.0
Khattak [62]	ML for Cyberattack Detection in V2I	Emissions increase (%), Fuel consumption impact	Smart Transportation, CDA

challenge, necessitating innovations such as lightweight policy updating, experience replay compression, and hardware-aware model tuning. Blockchain-based mechanisms stand out for their transparency and immutability, providing decentralized trust among distributed systems. Nevertheless, the energy burden of cryptographic operations and consensus protocols, such as Proof-of-Work (PoW), threatens their scalability in green infrastructure. The shift toward greener alternatives, Proof-of-Stake (PoS), Delegated PoS, and Directed Acyclic Graph (DAG) architectures, marks a crucial step toward reconciling blockchain's security benefits with sustainability imperatives. Meanwhile, ML and DL methods continue to deliver high detection accuracy and scalability across diverse applications, but often disregard the hardware-level optimizations necessary for low-power environments. Without such co-design between algorithms and embedded platforms, their ecological footprint remains disproportionately high. From a sectoral perspective, IoT and edge networks have achieved the highest maturity in integrating AI-based cybersecurity with sustainability considerations, followed by smart grids and industrial automation. In contrast, areas such as healthcare and water management lack empirical work on the lifecycle energy consumption, carbon emissions, and resource utilization of deployed models. This imbalance signals an urgent need for domain-specific frameworks that embed sustainability metrics into every stage of cybersecurity design, from model training to deployment and maintenance. Additionally, these insights underscore that sustainable cybersecurity cannot hinge on any single AI paradigm. Instead, the future lies in hybridized ecosystems that merge the strengths of multiple approaches: FL for privacy-preserving collaboration, DRL for intelligent and adaptive defense, and green blockchain for secure, low-carbon trust management. When designed cohesively, such frameworks have the potential to transform cybersecurity from an energy-intensive necessity into an environmentally conscious enabler of digital resilience. This convergence not only enhances system reliability and responsiveness but also aligns cybersecurity's evolution with the broader global pursuit of sustainability and responsible innovation.

6. Datasets

A systematic examination of the reviewed studies reveals diverse strategies for utilizing datasets aligned with specific cybersecurity domains and sustainability considerations. Table 6 provides a structured overview of the datasets employed across the reviewed studies, organized by their application domains. A prevalent pattern is the recurrent use of established network IDS datasets, e.g., CICIDS2017 [68], NSL-KDD [69], and UNSW-NB15 [70]. These datasets, prominently featured in studies such as Jamshidi et al. [48], Alsulami et al. [57], and Kalra et al. [52], provide network traffic data enhanced by diverse attack scenarios. Their widespread adoption facilitates benchmarking of IDS within IoT, cloud, and edge computing, particularly when resource constraints are critical. In the domain of energy and smart grid security, domain-specific datasets, e.g., the Irish Smart Energy Trials and the State Grid Corporation of China dataset, play a pivotal role, especially for electricity theft detection, as demonstrated by Takiddin et al. [51]. Similarly, datasets, e.g., Edge-IIoTset and WUSTL-IIoT-2021, used by Ahakonye et al. [65] support the evaluation of lightweight ML models in SCADA and IIoT security contexts. For FL and IoT-centric frameworks, datasets such as MNIST [71], CIFAR-10 [72], FEMNIST [73], and ImageNet [74] are frequently employed. However, they were initially developed for general classification tasks rather than cybersecurity applications. Studies such as Baqer et al. [37], Savoia et al. [35], and Qiu et al. [36] repurpose these datasets to evaluate the efficiency of distributed learning techniques under energy and communication constraints.

Several studies addressed the absence of suitable real-world datasets by generating synthetic and simulated data tailored to their specific use cases. For example, Khattak [62] employed VISSIM and autonomy to simulate smart transportation attacks, while Goyat et al. [46] generated synthetic

Table 6
Summary of Datasets Utilized in Reviewed AI-based Cybersecurity Studies

Study	Dataset	Application Domain
Jamshidi et al. [48], Alsulami et al. [57], Kalra et al. [52]	CICIDS2017	IoT/Edge IDS, Network Intrusion Detection
Ahakonye et al. [65]	Edge-IIoTset, WUSTL-IIoT-2021, ICS-SCADA	SCADA, IIoT Security
Takiddin et al. [51]	Irish Smart Energy Trials, State Grid of China	Smart Grid, Electricity Theft Detection
Qiu et al. [36], Savoia et al. [35]	CIFAR-10, ImageNet, FEMNIST, Speech Commands	FL for Security
Baqer et al. [37]	MNIST (Simulated IoT Networks)	FL for IoT
Maiwada et al. [49]	Kaggle DDoS Dataset	5G IDS, DDoS Mitigation
Durai et al. [64]	BODMAS, VirusTotal Malware Dataset	Malware Detection, Low-Carbon AI Training
Khattak [62]	VISSIM, Autonomie Traffic Simulations	Smart Transportation Security
Goyat et al. [46]	Simulated Green IoT Network Data	IoT Privacy and Security
Bushehri et al. [50]	Peer-to-Peer IoT Dataset (18,448 Samples)	IoT Energy Overconsumption Detection
Viegas et al. [56]	NSL-KDD	Embedded Systems IDS

IoT network data to explore privacy-preserving blockchain solutions.

7. Policy, Ethical, and Socio-economic Implications of Sustainable AI-based Cybersecurity

The integration of AI into cybersecurity, while advancing automation, threat detection, and system resilience, introduces complex ethical, policy, and socio-economic implications, especially when viewed through the lens of sustainability. As highlighted in Table 3, high-performance AI models such as DL and FL often entail substantial computational and energy costs, resulting in significant carbon emissions. This raises a fundamental ethical trade-off: the pursuit of robust cyber defense must not undermine environmental and social sustainability goals, including SDG 13 (Climate Action) and SDG 12 (Responsible Consumption and Production). The environmental footprint of AI-based security systems, if left unregulated, can counteract the broader objectives of a green digital transformation. From a policy perspective, there is an urgent need to establish governance frameworks that explicitly integrate sustainability metrics into cybersecurity standards and procurement practices. Governments and regulatory bodies should promote lifecycle-based assessments that quantify energy consumption, carbon emissions, and e-waste associated with AI-driven security infrastructures. Such evaluations could be formalized through sustainability audits or cybersecurity solution certification programs, analogous to existing environmental labeling in other technology sectors. These measures would incentivize the adoption of green AI practices and reinforce corporate accountability in line with Environmental, Social, and Governance (ESG) principles. Ethically, the sustainable design of AI-based cybersecurity also encompasses issues of fairness, transparency, and inclusivity. Energy-intensive AI models risk creating disparities between well-resourced organizations and those in developing economies, where computational capacity and access to renewable energy are limited. To mitigate this digital divide, ethical frameworks should encourage the development and open dissemination of lightweight, low-power models suitable for deployment on affordable and energy-constrained hardware. Moreover, transparency in algorithmic design and lifecycle reporting is vital to ensure that security benefits do not come at the cost of hidden environmental or social harm. The socio-economic implications extend beyond technical efficiency to workforce transformation, equitable access, and long-term resilience. Transitioning toward sustainable cybersecurity requires investments in green computing infrastructure, interdisciplinary education, and ethical AI literacy among practitioners and policymakers. Incentivizing public–private partnerships focused on sustainable innovation could also accelerate the integration of AI-based cybersecurity into national and global sustainability agendas. Ultimately, embedding sustainability considerations into cybersecurity policy and ethics frameworks ensures that technological progress aligns with ecological stewardship and social equity. In this way, AI-driven cybersecurity can evolve from a purely technical field into a cornerstone of sustainable digital governance.

7.1. Alignment of AI-based Cybersecurity with Sustainable Development Goals

Table 7 provides a structured mapping between AI-based cybersecurity solutions and specific United Nations SDGs, especially SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 13 (Climate Action). Table 7 underscores the tangible contributions of AI-driven cybersecurity solutions to the advancement of key SDGs. Primarily, SDG 7 (Affordable and Clean Energy) benefits from methodologies that enhance the security and operational efficiency of energy infrastructures, e.g., smart grids and energy-aware IoT networks. Studies, e.g., Jamshidi et al. [48] and Fraga-Lamas et al. [59], directly support the reliability and sustainability of energy systems through optimized IDS and edge AI applications. SDG 9 (Industry, Innovation, and Infrastructure) is widely addressed across the reviewed literature, reflecting the role of AI-based cybersecurity in fortifying industrial processes, IIoT ecosystems, and critical infrastructure against cyber threats while ensuring operational continuity. Meanwhile, SDG 13 (Climate Action) is increasingly recognized, as evidenced in works such as Qiu et al. [36] and Durai et al. [64], which explicitly evaluate carbon footprints and propose low-carbon AI training techniques. However, engagement with the SDGs

Table 7
Mapping of AI-based Cybersecurity Solutions to SDG Objectives

Study	SDG 7: Affordable and Clean Energy	SDG 9: Industry, Innovation, and Infrastructure	SDG 13: Climate Action
Jamshidi et al. [48]	Energy-efficient IDS for edge IoT in smart grids	Robust infrastructure for secure IoT	Reports carbon emissions of IDS deployments
Fraga-Lamas et al. [59]	Green IoT and Edge AI for energy optimization	Sustainable industrial IoT applications	Reduced carbon footprint through Edge AI
Qiu et al. [36]	FL in energy-aware applications	Enhancing distributed learning systems	Explicit carbon footprint analysis of FL
Durai et al. [64]	Low-carbon AI training models for energy systems	Efficient malware detection in industrial networks	Minimizing carbon emissions in AI training
Saxena et al. [53]	Secure and sustainable cloud data centers	Cloud infrastructure optimization	Energy and carbon reduction in data centers
Brudni et al. [60]	Optimizing antivirus energy use in AVs	Green security in automotive industry	Energy savings reduce environmental impact
Khattak [62]	-	Smart transportation infrastructure security	Quantifies emissions impacts of cyberattacks

remains uneven: while energy and industrial sustainability are well represented, climate impact assessments are less systematically integrated across cybersecurity research.

7.2. Cross-Domain Coverage of AI Methods and Sustainability

Figure 10 presents a systematic mapping of AI methods to application sectors and sustainability metrics as extracted from the reviewed studies. Figure 10 captures the frequency with which each AI technique is applied across various sectors, e.g, IoT, edge computing, smart grids, cloud/data centers, and smart transportation, alongside sustainability metrics such as energy efficiency and carbon emissions.

Figure 10 shows that FL and DRL are widely applicable

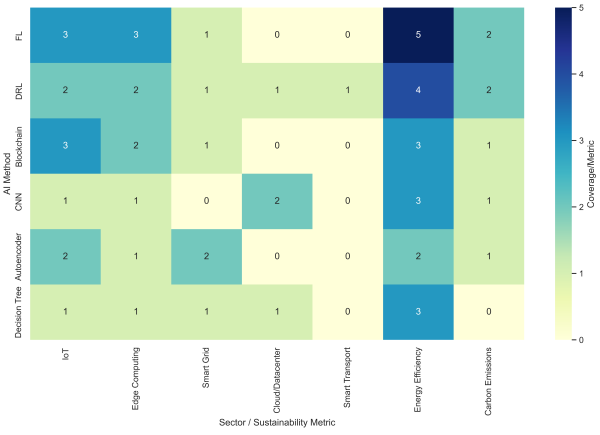


Figure 10: Coverage of AI methods across application sectors and sustainability metrics. The scale represents the number of studies addressing each intersection.

across multiple sectors, particularly in IoT and edge computing, and are strongly aligned with energy-efficiency metrics. Blockchain methods are widely adopted in IoT and smart grids, though their use in smart transportation and carbon emission assessments remains limited. CNNs and AEs are predominantly used in smart grids and cloud/data centers, often for anomaly detection and energy efficiency. Despite these advancements, a clear imbalance remains in the coverage of sustainability metrics. Energy efficiency is the most widely addressed metric across AI methods, whereas explicit carbon-emissions assessments are underrepresented. Furthermore, the smart transportation and healthcare sectors appear less explored in terms of integrating AI-based cybersecurity with sustainability considerations.

8. The Imperative for Standardized Sustainability Metrics in AI-based Cybersecurity

As AI-based cybersecurity solutions proliferate across diverse sectors, integrating sustainability principles into their evaluation remains inconsistent and underdeveloped. A critical barrier to holistic assessment is the absence of standardized key performance indicators that quantify the environmental and resource impacts of these technologies. Without unified metrics, it is challenging to compare models across studies, gauge their real-world sustainability, or align them with broader environmental objectives, such as the SDGs. To bridge this gap, we propose a set of standardized key performance indicators (KPIs) presented in Table 8 that can serve as a foundational framework for future research and development in sustainable AI-driven cybersecurity. These indicators extend beyond direct energy consumption and carbon emissions to encompass computational efficiency, hardware longevity, water footprint, thermal output, and material recyclability. Such dimensions are critical for deployment in energy-constrained, resource-limited, and environmentally sensitive environments, including IoT networks, edge computing, and large-scale cloud infrastructures. Standardizing these key performance indicators would not only enhance cross-study comparability but also enable the cybersecurity community to systematically address environmental, energy, and resource considerations alongside security performance. Incorporating these indicators facilitates the development of secure, resilient, and environmentally responsible digital infrastructures, contributing directly to SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 13 (Climate Action).

Table 8
Proposed Standardized Sustainability Key Performance Indicators for AI-based Cybersecurity

Sustainability Metric	Description and Significance
Energy Consumption (Wh)	Measures the total energy used during training, inference, and system operation. Essential for understanding operational efficiency, particularly in IoT, edge computing, and embedded systems, where energy budgets are critical.
Carbon Emissions (kg CO_2e)	Calculates the greenhouse gas emissions attributable to energy consumption, considering regional grid carbon intensities. Supports alignment with climate-focused objectives such as SDG 13 (Climate Action).
Model Efficiency (Inference Time vs. Power Consumption)	Captures the trade-off between processing speed and energy use during inference. Vital for evaluating real-time systems where latency and power draw must be balanced.
Resource Utilization (CPU/GPU Cycles)	Quantifies the computational demands on processing units, reflecting the scalability and environmental footprint of deploying AI models on different hardware platforms.
Hardware Longevity (Years)	Assesses the expected operational lifespan of hardware when running AI models. Prolonging hardware use reduces electronic waste and the environmental impact of manufacturing new components.
Water Footprint (Liters)	Estimates the water consumption involved indirectly through energy production, especially in regions where power generation is water-intensive. Relevant for comprehensive sustainability assessments.
Material Recyclability (%)	Evaluates the proportion of hardware materials that can be recycled after end-of-life. Supports circular economy principles and reduces landfill waste.
Thermal Output (BTU/h)	Measures the heat generated by AI systems during operation. A higher thermal output can increase cooling requirements, thereby impacting overall energy efficiency.
Memory Footprint (GB)	Captures the memory requirements of AI models during inference and training. Lower memory usage can enhance deployment feasibility on resource-constrained devices.
Scalability Efficiency (Performance per Additional Node)	Assesses how efficiently a model scales across distributed systems, balancing performance gains with resource overheads. Critical for cloud-based cybersecurity deployments.

9. Comparative Trade-offs Between AI Techniques in Security and Sustainability

Table 9 illustrates that no single AI technique simultaneously optimizes both security performance and sustainability metrics. DRL and blockchain-based methods demonstrate high security efficacy but suffer from poor carbon awareness and high resource demands, making them less viable in energy-sensitive deployments. FL offers a balanced approach, enhancing privacy and providing moderate energy and resource efficiency, yet still incurs observable carbon costs. LLMs, though emerging in cybersecurity, exhibit significant sustainability drawbacks due to intensive training energy and resource requirements, compounded by a current lack of targeted studies. In contrast, Lightweight ML and spiking neural networks show promise for sustainable cybersecurity, offering high to very high energy efficiency and low resource utilization, although often at the expense of trade-offs in security performance and the maturity of the application. Beyond qualitative trade-offs, model complexity plays a decisive role in determining the practical feasibility of AI techniques for sustainable cybersecurity. Model complexity can be characterized through several dimensions, including parameter count, floating-point operations (FLOPs) [75], inference latency, memory footprint, and hardware dependency. DRL and large language models (LLMs) typically involve millions to billions of parameters, resulting in high FLOPs and substantial memory requirements during both training and inference. These characteristics significantly increase energy consumption and latency, rendering such models challenging to deploy on edge devices, IoT nodes, and embedded systems with limited computational resources. Additionally, reliance on GPUs or specialized accelerators further limits their applicability in decentralized and energy-sensitive environments.

FL exhibits intermediate complexity characteristics. While individual client models may be lightweight, the cumulative computational and communication overhead across multiple training rounds increases overall energy usage and carbon footprint. Communication frequency, model size, and aggregation strategies directly impact FLOPs and latency, indicating that FL scalability is highly sensitive to model complexity and network conditions. Without careful model compression or adaptive aggregation, FL-based security solutions may still exceed the resource limits of large-scale edge deployments. In contrast, lightweightML models and spiking neural networks demonstrate significantly lower complexity profiles. These approaches typically involve fewer parameters, reduced FLOPs, and predictable inference latency, enabling deployment on low-power CPUs and microcontrollers. Their low memory footprint and hardware compatibility make them particularly suitable for real-time intrusion detection in resource-constrained environments. However, this reduction in complexity often comes at the cost of limited representational capacity and reduced adaptability to sophisticated attack patterns. Consequently, complexity-aware design emerges as a critical factor in balancing security effectiveness, sustainability, and deployability. The reviewed literature reveals that model complexity is closely tied to energy efficiency, carbon impact, and deployment feasibility, yet it is rarely explicitly quantified. The lack of standardized reporting on parameters, FLOPs, latency, and hardware requirements hinders reproducibility and prevents meaningful comparison across studies. Addressing this gap requires future work to integrate complexity-aware evaluation metrics alongside security and sustainability indicators, particularly for edge-centric and real-time cybersecurity applications.

Table 9
Comparative Trade-offs Among AI Techniques in Cybersecurity and Sustainability

AI Technique	Security Performance	Energy Efficiency	Carbon Awareness	Resource Utilization	Example Studies
DRL	High	Moderate	Low	High	[48]
FL	High (Privacy)	Moderate	Moderate	Moderate	[35], [36]
Blockchain-based Methods	High (Decentralization)	Low	Low	High	[42], [46]
LLMs	Emerging	Low (Training)	Low	High	(Gap in literature)
Lightweight ML	Moderate	High	Low	Low	[56], [59]
Spiking Neural Networks	Moderate	Very High	Low	Very Low	(Potential for future research)

10. Toward a Unified Sustainability-Aware Cybersecurity Design Framework

Despite the growing body of research evaluating AI-driven cybersecurity systems in energy-aware, resource-constrained settings, the literature lacks a unified theoretical formulation that jointly models security effectiveness and sustainability constraints. Existing works primarily report isolated performance metrics such as detection accuracy, energy consumption, and carbon emissions independently. However, secure and sustainable digital infrastructures require a mathematically grounded co-optimization perspective. In this section, we formalize the sustainability-aware cybersecurity design problem and introduce a unified evaluation metric to enable standardized comparison across heterogeneous AI paradigms.

10.1. Multi-Objective Security–Sustainability Optimization Model

Let $\mathcal{M}$ denote an AI-driven cybersecurity model deployed in a digital infrastructure. We define security performance as a composite function:

$$S(\mathcal{M}) = \omega_1 Acc + \omega_2 F1 + \omega_3 Recall + \omega_4 Robustness \quad (1)$$

The function $S(\mathcal{M})$ aggregates multiple security-relevant performance indicators into a single scalar objective. Here, Acc represents overall detection accuracy, $F1$ captures the harmonic balance between precision and recall, $Recall$ reflects the system's sensitivity to malicious events, and $Robustness$ measures resistance against adversarial perturbations and zero-day attacks. The weighting coefficients ω_i allow domain-dependent prioritization. For instance, in critical infrastructure protection, recall may be assigned a higher weight to minimize false negatives, whereas in large-scale cloud systems, robustness against adversarial drift may dominate. Energy efficiency is defined as energy per inference:

$$E(\mathcal{M}) = \frac{E_{total}}{N_{inference}} \quad (2)$$

This formulation normalizes total energy consumption by operational workload, ensuring comparability across heterogeneous systems. E_{total} includes computational energy consumed during forward propagation, memory access, and I/O operations. Dividing by the number of inferences enables fair benchmarking between lightweight edge models and large centralized architectures. Lower $E(\mathcal{M})$ indicates improved operational sustainability. Lifecycle carbon emission is modeled as:

$$C(\mathcal{M}) = \sum_{k=1}^{L} E_k \cdot \kappa_k \quad (3)$$

where E_k denotes energy consumption at lifecycle stage k, including training, deployment, communication, maintenance, and hardware aging. The coefficient κ_k represents carbon intensity ($kgCO_2e$ per Joule) associated with the energy source at stage k. This formulation explicitly captures geographical and infrastructural variability, since carbon intensity differs across regions and energy mixes. Unlike inference-only measurements, this lifecycle-aware model accounts for hidden environmental costs such as retraining cycles and federated communication overhead. Resource utilization is expressed as:

$$R(\mathcal{M}) = \alpha \cdot CPU_{util} + \beta \cdot MEM_{util} + \delta \cdot BW_{util} \quad (4)$$

This term quantifies system-level strain imposed by the model. CPU_{util} reflects processor occupancy, MEM_{util} captures memory pressure, and BW_{util} represents communication bandwidth usage. The coefficients α, β, and δ enable infrastructure-specific calibration, since resource scarcity varies between edge IoT devices and cloud environments. High $R(\mathcal{M})$ values indicate potential degradation of co-located services and reduced system longevity. The sustainability-aware cybersecurity design problem can therefore be formulated as:

$$\max_{\mathcal{M}} S(\mathcal{M}) \quad (5)$$

subject to:

$$E(\mathcal{M}) \le E_{max} C(\mathcal{M}) \le C_{budget} R(\mathcal{M}) \le R_{threshold} \quad (6)$$

The objective seeks to maximize security effectiveness while respecting sustainability constraints. E_{max} defines acceptable operational energy limits, C_{budget} imposes carbon emission boundaries aligned with environmental policy targets, and $R_{threshold}$ prevents excessive infrastructure stress. This formulation transforms cybersecurity design into a constrained multi-objective optimization problem. Instead of optimizing security in isolation, the solution space is bounded by ecological and infrastructural feasibility. The

optimal configurations lie on a Pareto frontier, defined as the set of models for which no improvement in $S(\mathcal{M})$ can be achieved without increasing $E(\mathcal{M})$, $C(\mathcal{M})$, or $R(\mathcal{M})$. Operating on this frontier ensures balanced trade-offs between robustness and environmental responsibility, enabling the systematic co-design of secure, sustainable AI-driven systems.

10.2. GreenSec Index: A Unified Sustainability Metric

To enable standardized benchmarking across AI paradigms, we introduce the Green Security Index (GSI), defined as:

$$GSI = \frac{S(\mathcal{M})^{\gamma}}{E(\mathcal{M})^{\lambda} \cdot C(\mathcal{M})^{\mu}} \tag{7}$$

where γ, λ, and μ are tunable weights reflecting deployment context. Higher GSI values indicate superior security-to-sustainability efficiency. The GSI metric enables direct comparison between DRL, FL, blockchain-based security, lightweight ML, and emerging LLM-driven cybersecurity approaches under a unified scale.

10.3. Lifecycle-Aware Sustainability Modeling

Most prior research focuses exclusively on inference-time energy, neglecting lifecycle impact. We define total lifecycle carbon impact as:

$$C_{lifecycle} = C_{training} + C_{deployment} + C_{communication} + C_{maintenance} \tag{8}$$

This lifecycle-aware modeling reveals that communication-intensive paradigms such as Federated Learning may exhibit lower inference energy but significantly higher communication emissions, while blockchain-based systems incur substantial cryptographic overhead. In contrast, lightweight ML models have a lower lifecycle impact due to minimal retraining and lower hardware stress.

10.4. Pareto Trade-Off Analysis

Security and sustainability objectives inherently conflict. Increasing model depth improves robustness but raises energy consumption. Decentralization enhances privacy but increases communication emissions. Formally, the trade-off surface can be expressed as:

$$\begin{aligned} \mathcal{P} = \{(S, E, C, R) \mid \nexists \mathcal{M}' \text{ such that} \\ S' \geq S \ \wedge \ E' \leq E \ \wedge \ C' \leq C \ \wedge \ R' \leq R\} \end{aligned} \tag{9}$$

This Pareto [76] set defines optimal sustainability-aware cybersecurity configurations. Future systems should operate along this frontier rather than optimizing security alone.

10.5. Policy and SDG Alignment Formalization

To align cybersecurity design with global sustainability goals, we define a policy compliance constraint:

$$\Phi(\mathcal{M}) = f(SDG_7, SDG_9, SDG_{12}, SDG_{16}) \tag{10}$$

where $\Phi(\mathcal{M})$ evaluates whether deployment satisfies clean energy usage, resilient infrastructure support, responsible consumption, and secure institutions. Embedding policy constraints into the optimization problem ensures that AI-driven cybersecurity systems are not only technically efficient but also socially and environmentally aligned. To illustrate how different AI paradigms position themselves within the proposed sustainability-aware cybersecurity framework, Table 10 provides a qualitative comparative assessment across key dimensions, including security strength, energy efficiency, carbon impact, and GSI potential. The table synthesizes findings from the reviewed literature and maps each methodological family onto the proposed unified evaluation space.

11. Discussion

The convergence of AI-driven cybersecurity and sustainability presents complex technical challenges that require holistic optimization across algorithmic efficiency, system architecture, and environmental stewardship. The reviewed body of work illustrates significant advancements in model optimization and resource-aware design, yet it reveals persistent gaps in standardization, careful evaluation, and cross-domain applicability. Techniques such as DRL and FL have been particularly effective in reducing energy consumption while preserving security performance, as evidenced by Jamshidi et al. [48] and Savoia et al. [35], who demonstrated adaptive IDS models and client selection strategies for edge environments. These approaches dynamically adjust learning and detection policies in response to fluctuating energy budgets and computational constraints. However, they are often evaluated in isolated testbeds without consideration for lifecycle carbon impacts and scalability across diverse infrastructure types.

In blockchain-oriented cybersecurity, efforts by Aldawsari et al. [42] and Ahmed et al. [43] integrate decentralized trust mechanisms with energy-aware clustering and routing. However, blockchain's inherent computational overhead, particularly in consensus protocols, is seldom quantified in terms of emissions per transaction and energy cost per secured packet[77][78][79]. Similarly, while lightweight cryptographic protocols and clustering optimizations (e.g., Gurram et al. [41], Rehman et al. [45]) reduce localized energy expenditure, assessments of end-to-end environmental costs remain absent.

Moreover, the scarcity of datasets annotated with sustainability parameters (such as energy per inference, carbon emissions per model epoch) limits the ability to evaluate trade-offs between cybersecurity robustness and environmental impact holistically. Current reliance on datasets such as CICIDS2017, NSL-KDD, and Edge-IIoTset ([48], [56], [65]) primarily benchmarks detection accuracy, leaving a critical gap in performance assessment under resource constraints. This gap is compounded by the absence of standardized sustainability key performance indicators (KPIs)

Table 10
Comparative Sustainability Positioning of AI-Based Cybersecurity Paradigms

Method	Security Strength	Energy Efficiency	Carbon Impact	GSI Potential
DRL	High	Moderate	Moderate	High
FL	High	Moderate	High	Moderate
Blockchain-Based Security	Moderate	Low	Very High	Low
Lightweight ML (DT/RF)	Moderate	High	Low	High
LLM-Based Security	Very High	Low	Very High	Low

that encapsulate energy consumption (Wh), carbon emissions (kg CO_2e), computational efficiency (e.g., FLOPs per inference), and hardware resource utilization (CPU/GPU cycles). The reviewed methodologies also reveal limited cross-layer optimizations, where software-level AI models are co-designed with hardware acceleration (e.g., FPGA implementations, neuromorphic chips), despite promising outcomes in hardware-aware energy savings, as demonstrated by Shafique et al. [61] and Viegas et al. [56]. This underlines an urgent need for integrated design frameworks that align algorithmic choices with energy profiles and hardware capabilities.

Additionally, sectoral applications remain uneven. Although smart grids, IoT, and edge computing receive considerable attention, critical sectors such as healthcare and smart transportation (e.g., [62]) are underexplored, particularly in contexts that require the stringent co-optimization of safety and sustainability. A broader, domain-specific adaptation of AI security models, tailored to unique resource constraints and regulatory requirements, remains an open research frontier.

Furthermore, the comparative synthesis highlights that while most existing reviews analyze AI for cybersecurity or AI for sustainability independently, this study uniquely integrates both through a unified conceptual framework. The proposed framework demonstrates its practical usefulness by enabling a structured comparison of studies under common sustainability metrics (energy, carbon, and computational efficiency) and by mapping observed trade-offs into actionable research directions. Although a full-scale case study lies beyond the scope of this paper, the conceptual demonstration in Section **??** illustrates how sustainability-aware AI architectures can be evaluated using multi-objective performance criteria. This demonstrates the framework's potential for real-world adaptation across various sectors, including smart grids and industrial IoT.

The discussion also acknowledges methodological limitations that influence generalization. The analysis relies primarily on published literature, which may introduce publication bias and underrepresent commercial or proprietary solutions. Moreover, while the sustainability metrics were consistently referenced, inconsistent reporting across studies restricted full quantitative comparison. Future work should apply the framework to empirical case studies, integrating standardized measurement tools and datasets co-labeled for security and sustainability to validate its utility across industrial contexts.

Moreover, these observations underscore the need to establish standardized sustainability key performance indicators, develop multi-objective optimization frameworks that jointly address security and sustainability, and advance policy guidelines that mandate environmental impact assessments for AI-based cybersecurity solutions. Such integrative strategies are critical to realizing AI security systems that are not only resilient and robust but also environmentally responsible and aligned with the SDGs.

A critical analysis of trade-offs across the reviewed literature reveals that while deep and transformer-based IDS models achieve superior detection accuracy, they impose high energy and carbon costs that directly conflict with sustainability goals. In contrast, lightweight and hybrid AI methods, such as decision trees, FL, and RL, offer a more favorable balance between cybersecurity robustness and energy efficiency. This contradiction underscores the need for adaptive strategies that align algorithmic complexity with sustainability metrics. Moreover, open debates persist regarding the optimal balance between detection fidelity and environmental impact, highlighting an unresolved research frontier that demands cross-layer co-optimization of models, data, and hardware. Beyond technical contributions, the findings of this review have important practical implications for industry, policymakers, and system designers. For practitioners, deploying green AI in cybersecurity should prioritize lightweight architectures, adaptive inference scheduling, and lifecycle carbon accounting to ensure efficient operation without compromising detection accuracy. System designers are encouraged to integrate energy benchmarking and model transparency tools during the development phase to strike a balance between accuracy and resource efficiency. For policymakers, aligning cybersecurity standards with environmental regulations and sustainability reporting (e.g., carbon disclosure and green AI certification) will be key to promoting accountable digital infrastructures. Moreover, recent works on green supply chain management for digital assets in the Metaverse and sustainable, secure communication for Industry 5.0 electric vehicle ecosystems highlight cross-sector synergies that future cybersecurity frameworks should leverage.

11.1. Analysis of Sustainability Priorities in Cybersecurity Design

Figure 11 presents a structured, multidimensional comparison across three critical axes of sustainable cybersecurity: Technical, Economic, and Ethical. The technical dimension scores a full 5/5 across all evaluation criteria, design priority, evaluation consideration, and literature coverage, reflecting the dominant emphasis on performance, detection accuracy, and system resilience in current security research. In contrast, the Economic and Ethical dimensions exhibit significant under-representation, with declining scores, particularly in literature coverage (2/5 each). Economic aspects such as energy consumption, carbon footprint, and infrastructure costs are often overlooked, while ethical concerns, including fairness, transparency, and privacy, are rarely operationalized in system design. This imbalance reveals a fundamental gap in integrating sustainability-oriented objectives into cybersecurity frameworks.

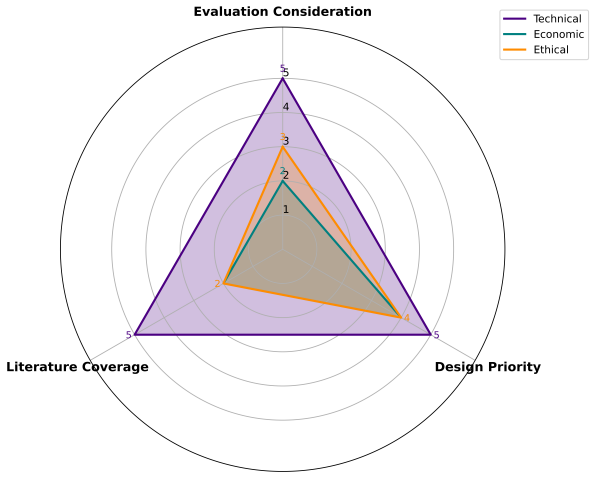


Figure 11: Comparison of Sustainable Cybersecurity Dimensions: Technical, Economic, and Ethical Perspectives. The figure illustrates the relative emphasis on three design axes across the criteria of design priority, evaluation consideration, and literature coverage.

11.2. Privacy Preservation versus Carbon Footprint

Privacy-preserving cybersecurity mechanisms introduce a second critical trade-off between data protection and environmental impact. Techniques such as FL, secure aggregation, and encrypted model updates are designed to reduce data exposure and enhance compliance with privacy regulations. While these approaches successfully decentralize sensitive data, they impose additional communication rounds, cryptographic operations, and synchronization overhead, which collectively increase energy consumption and carbon emissions. Several studies report that FL frameworks can generate higher cumulative emissions than centralized training when client participation is frequent, and communication efficiency is not optimized. This finding contradicts the common assumption that privacy-preserving approaches are inherently more sustainable. Without energy-aware client selection, adaptive communication scheduling, and carbon-aware aggregation strategies, privacy enhancement may directly conflict with sustainability objectives. These observations underscore the need for co-optimization frameworks that consider privacy and environmental impact as coupled design constraints, rather than independent goals.

11.3. Accuracy versus Energy Consumption

A central contradiction identified across the reviewed literature lies in the trade-off between detection accuracy and energy efficiency. DL and transformer-based intrusion detection systems consistently report superior detection accuracy and robustness against sophisticated attack patterns, particularly in high-dimensional traffic and encrypted environments. However, this performance gain is achieved through increased model depth, parameter count, and computational complexity, which directly translate into higher energy consumption during both the training and inference phases. Empirical studies show that continuous retraining and real-time inference in DL-based systems significantly increase power usage, making such approaches difficult to sustain in edge and IoT deployments with strict energy budgets. In contrast, lightweightML models, including decision trees, shallow neural networks, and hybrid statistical methods, exhibit reduced detection accuracy in highly adversarial scenarios but consume substantially less energy and offer faster inference. These models are more compatible with long-term deployment in resource-constrained environments, albeit at the expense of marginal detection performance. This contradiction highlights that optimizing exclusively for accuracy often yields energy-inefficient solutions, while prioritizing energy efficiency may compromise detection fidelity. The absence of standardized metrics that jointly quantify accuracy and energy cost further complicates informed design decisions.

11.4. Security Strength versus Latency and Responsiveness

A further contradiction arises between security strength and system latency. Advanced AI-based security models, particularly deep neural networks and ensemble architectures, improve detection robustness and reduce false positives. However, increased model complexity often leads to higher inference latency, which is problematic for time-sensitive applications such as industrial control systems, smart grids, and vehicular networks. Elevated latency not only degrades system responsiveness but may also introduce safety risks in mission-critical environments. Lightweight and rule-based security mechanisms offer faster response times and predictable latency profiles, albeit with reduced adaptability to novel attack patterns. Blockchain-enabled security frameworks exacerbate latency challenges due to he overhead of consensus protocols and transaction validation. These findings indicate that security strength and responsiveness are frequently at odds, particularly in distributed and real-time systems. Balancing this trade-off requires adaptive architectures that dynamically adjust model

complexity and security depth in response to latency constraints and operational context.

11.5. Limitations

Despite significant advancements in AI-driven cybersecurity solutions, the integration of sustainability remains partial and fragmented, as delineated in Table 11. A key limitation is the narrow interpretation of sustainability, where most studies primarily address energy efficiency without encompassing broader environmental dimensions, e.g., carbon emissions, resource depletion, and lifecycle ecological impacts. For instance, works, e.g., Savoia et al. [35] and Jamshidi et al. [48] achieve significant energy optimizations in FL and RL contexts; however, they lack environmental assessments, including carbon footprint and e-waste generation, both of which are critical for sustainable cybersecurity solutions.

Another major constraint is the scarcity of datasets that combine cybersecurity threat data with sustainability metrics such as power consumption, latency, and carbon intensity. Studies such as Alsulami et al. [57] and Takiddin et al. [51] rely on conventional IDS, which omit sustainability-relevant annotations. This disconnect hampers the development and benchmarking of AI models optimized for both security and environmental sustainability.

Blockchain-based security solutions, as explored by Aldawsari et al. [42] and Ahmed et al. [43], introduce significant energy overhead due to the computation-intensive nature of consensus mechanisms. The absence of research into greener blockchain alternatives, such as DAG-based protocols, poses a sustainability challenge that remains largely unaddressed.

Hardware-software co-optimization is an area that has been insufficiently explored, which limits the deployment of sustainable AI solutions in edge and embedded systems. Only a few works, such as Shafique et al. [61] and Viegas et al. [56], leverage techniques, e.g., hardware acceleration, model pruning, or neuromorphic computing to balance security performance with sustainability goals.

From a sectoral perspective, research predominantly targets IoT, smart grids, and edge environments, with limited exploration of sectors, e.g., healthcare, industrial control systems, and smart transportation ([62], [63]), where sustainability constraints are equally critical. Compounding these issues is the absence of standardized sustainability key performance indicators in cybersecurity research. Without unified metrics such as energy consumption per inference (Wh), carbon emissions per model operation (kg CO_2e), and resource utilization rates (CPU/GPU cycles), cross-study comparability and holistic sustainability assessments remain elusive ([53], [60]). Additionally, ethical and policy dimensions are underdeveloped. Few studies explicitly address the environmental cost of deploying energy-intensive AI models for security tasks and propose frameworks for regulatory compliance aligned with the SDGs. Works, such as those by Brudni et al. [60] and Saxena et al. [53], initiate this discourse; however, comprehensive ethical guidelines and policy-driven sustainability mandates are still lacking.

Beyond these thematic limitations, several methodological gaps also constrain this review. First, the analysis is based predominantly on published academic studies, which may introduce publication bias and underrepresent industrial implementations or proprietary AI-driven cybersecurity frameworks. Second, while sustainability metrics, energy consumption (J/inference), CO_2 emissions per epoch, and operations per Watt, were applied in the qualitative synthesis, heterogeneous reporting across studies limited quantitative comparability. Third, most of the analyzed datasets are simulation-based and derived from laboratory environments, rather than large-scale real-world deployments, which restricts the generalizability of the findings.

12. Challenges and Gaps

To address RQ3, this systematic review identifies critical gaps and challenges at the intersection of AI-driven cybersecurity and sustainability. While advances in detection accuracy and decentralized trust mechanisms are evident, the integration of sustainability remains fragmented. As summarized in Table 12, unresolved challenges span the lack of standardized metrics, energy-intensive blockchain operations, limited datasets, and weak policy integration. The following subsections synthesize these key gaps, focusing on the most critical and distinct issues that hinder the practical realization of sustainable cybersecurity systems.

12.1. Lack of Standardized Sustainability Metrics

The absence of standardized sustainability indicators across AI-driven cybersecurity studies remains a core limitation. Most works, such as Jamshidi et al. [48] and Saxena et al. [53], emphasize detection performance metrics (accuracy, recall, and precision) while neglecting quantifiable measures, including energy consumption, CO_2 emissions, and computational load. This inconsistency impedes cross-study benchmarking and obscures environmental costs. Qiu et al. [36] provide a rare evaluation of CO_2 within FL contexts, highlighting the need for unified sustainability KPIs, such as energy per inference, carbon intensity, and hardware utilization efficiency. Extending benchmarking frameworks such as MLPerf or OpenML with sustainability modules could standardize such reporting, promoting reproducible and environmentally accountable AI security evaluations.

12.2. Energy-Intensive Blockchain Security Solutions

Blockchain-based cybersecurity frameworks, including those by Aldawsari et al. [42] and Ahmed et al. [43], ensure transparency and decentralization but remain hindered by the energy demands of consensus algorithms such as Proof-of-Work (PoW). These architectures are impractical for constrained IoT devices and conflict with SDG 7 (Affordable and Clean Energy). Transitioning toward green blockchain mechanisms—such as Proof-of-Stake (PoS), Delegated PoS, or Directed Acyclic Graph (DAG)—and hybrid on/off-chain designs could mitigate this problem. Integrating adaptive

Table 11
Summary of Key Limitations in AI-based Cybersecurity Solutions with Sustainability Considerations

Limitation Aspect	Description and Representative Studies
Narrow Sustainability Scope	Focus on energy consumption without holistic environmental metrics (carbon emissions, lifecycle impact). Examples: [35], [48].
Dataset Limitations	Lack of datasets annotated with both threat and sustainability metrics (energy, carbon, resource usage). Affects: [57], [51], [50].
Blockchain Energy Overhead	High energy cost of consensus algorithms; few explorations of green blockchain alternatives. Examples: [42], [43].
Limited Hardware-Software Co-Optimization	Sparse adoption of energy-efficient hardware optimizations (e.g., quantization, neuromorphic chips). Notable works: [61], [56].
Sectoral Underrepresentation	Dominance of IoT and smart grids, with limited research in healthcare, ICS, smart transportation. Noted by: [62], [63].
Lack of Standardized key performance indicator	Absence of unified metrics for sustainability in cybersecurity (e.g., Wh, CO_2e, CPU/GPU cycles). Highlighted by: [53], [60].
Ethical and Policy Gaps	Insufficient frameworks addressing ethical trade-offs and policy compliance for sustainable AI security. Initiated by: [60], [53].

energy-aware scheduling can further optimize trust and security while minimizing carbon impact, making blockchain applications more feasible in sustainable IoT deployments.

12.3. Inadequate Datasets Combining Security and Sustainability

Data limitations remain a major barrier. As noted by Ahakonye et al. [65], existing SCADA and IIoT datasets lack joint annotations for security threats and sustainability parameters (e.g., latency, energy consumption, emissions). This separation limits dual-objective evaluation and model optimization. Developing multimodal datasets that integrate both aspects would enable more balanced analyses of performance and sustainability. Collaborative, open-source initiatives involving academia, industry, and government should focus on dataset standardization, inclusion of diverse attack types, and reproducible measurement of environmental impact to support comprehensive benchmarking.

12.4. Weak Hardware-Software Co-Design Strategies

Viegas et al. [56] demonstrated that hardware-aware modeling reduces IDS energy consumption, yet broader hardware-software co-design remains rare. Many AI models are built without considering the constraints of low-power or embedded environments, resulting in inefficiencies when deployed outside laboratory conditions. Integrating neuromorphic computing and energy-aware neural architecture search could address this gap. Co-design frameworks that simulate algorithmic and hardware-level interactions before deployment would ensure optimal energy efficiency, particularly for edge-based AI security models aligned with SDG 9 (Industry, Innovation, and Infrastructure).

12.5. Resource Overhead in Edge-Based Security Models

Although DRL-based security systems, such as those by Jamshidi et al. [23], enhance responsiveness and adaptability, they often incur excessive energy and memory overhead on IoT devices. These limitations undermine sustainability and scalability. Model compression techniques, such as pruning, quantization, and knowledge distillation, can reduce energy consumption without significant accuracy losses. Moreover, employing lightweight LLMs for selective inference or adaptive compression pipelines that adjust model complexity in real time based on resource conditions could preserve detection performance under tight energy constraints.

12.6. Sectoral Bias in Research Focus

Current research disproportionately targets IoT and edge networks [46], while critical sectors, such as healthcare and smart transportation, remain understudied despite their high relevance to sustainability. Khattak [62] highlights progress in vehicular networks, yet energy-efficient healthcare security frameworks remain scarce. Expanding AI-driven sustainable security research into these domains, with domain-specific benchmarks for energy and carbon optimization, would enhance cross-sector resilience and ensure equitable integration of sustainability.

12.7. Omission of Lifecycle Carbon Accounting

Lifecycle carbon assessment across AI models remains neglected. Although Qiu et al. [36] measured CO_2 emissions during training, few works evaluate the cumulative footprint from training to inference. This oversight limits accountability and masks long-term environmental costs, especially for large-scale deployments. Incorporating lifecycle assessment frameworks into cybersecurity evaluations would align research with corporate sustainability standards and SDG 13 targets. Open-source lifecycle toolkits for AI-based cybersecurity could facilitate transparent tracking of emissions and enable data-driven design choices.

12.8. Limited Interpretability of AI Security Models

Complex AI architectures, such as DRL [48] and AEs [51], often operate as black boxes, obscuring both decision logic and sustainability trade-offs. This lack of transparency reduces trust in critical domains such as smart grids and healthcare. Explainable AI (XAI) integrated with energy profiling could bridge this gap, enabling stakeholders to interpret decisions in conjunction with energy costs. For instance, an explainable DRL IDS reporting energy per detection [48] would allow practitioners to balance performance and resource efficiency. Standardized interpretability

Table 12
Synthesized Challenges, Implications, and Research Pathways for Sustainable AI-Driven Cybersecurity

Challenge	Implications and Study References	Research Directions and Solutions
Lack of Standardized Sustainability Metrics	Inconsistent reporting of energy consumption, carbon emissions, and resource utilization hinders comparability across studies [36, 48, 53]. Without KPIs, aligning with SDG 13 (Climate Action) is challenging.	Establish unified sustainability KPIs (e.g., Joules per inference, CO_2 per epoch, hardware-performance ratios) and adopt these as standard benchmarks in security research.
Energy-intensive blockchain Security Solutions	Blockchain-based approaches enhance trust but often rely on high-energy consensus protocols (e.g., PoW) unsuited for constrained IoT devices [42, 43].	Transition to green blockchain mechanisms (e.g., PoS, DPoS, DAG) and integrate energy-aware scheduling for secure, low-carbon operation.
Insufficient Security-Sustainability Integrated Datasets	Lack of datasets combining security threats with sustainability metrics (energy, latency, emissions) limit dual-objective optimization [65].	Create multimodal benchmark datasets with parallel attack labels and sustainability profiles for holistic evaluation.
Weak Hardware-Software Co-Design Strategies	Models are developed without hardware profiling, leading to sub-optimal performance on low-power devices [56].	Use co-design strategies with neuromorphic computing, hardware-aware neural architecture search, and deployment-optimized training.
Resource Overhead in Edge-Based Security Models	DRL and deep IDS models consume excessive compute and memory in IoT settings [23].	Apply pruning, quantization, knowledge distillation, and efficient LLMs for real-time, low-energy inference.
Sectoral Bias in Research Focus	Overemphasis on IoT leaves healthcare, transportation, and other critical sectors underexplored [46, 62].	Expand sustainability-aware security models to diverse domains using tailored benchmarks and constraints.
Omission of Lifecycle Carbon Accounting	Energy use in training is sometimes measured, but deployment and inference phases are ignored [36].	Integrate lifecycle carbon accounting into evaluation pipelines to capture full environmental impact.
Limited Interpretability of AI Security Models	Black-box models obscure resource trade-offs and hinder trust in critical infrastructure [48, 51].	Develop explainable AI with embedded energy/cost profiling to make security–sustainability trade-offs transparent.
Inconsistent Model Adaptation to Resource Constraints	Static models fail to adjust under fluctuating CPU, battery, or bandwidth conditions [41].	Implement adaptive models via meta-learning and RL with real-time resource feedback loops.
Absence of Multi-Objective Optimization Frameworks	Most works optimize security or sustainability separately, missing Pareto-optimal trade-offs [35].	Use evolutionary algorithms and Pareto optimization to jointly maximize detection, energy efficiency, and low latency.
Scarcity of Real-Time Energy-Aware AI Security Models	Models rarely self-adjust based on live energy measurements [49].	Embed energy sensors and adaptive scheduling to match computational load to available resources dynamically.
Environmental Cost of Model Retraining	Frequent retraining of large models causes high emissions and costs [48].	Employ continual FL, transfer learning, and adaptive retraining triggers based on threat intelligence and sustainability thresholds.
Insufficient Cross-Layer Optimization	Efficiency gains at one layer are negated by inefficiencies in hardware or networking [61].	Apply holistic optimization across AI models, communication protocols, and hardware platforms.
Underutilization of LLMs for Sustainability-Aware Security	LLMs are rarely optimized for low-resource, secure environments.	Design lightweight LLMs via pruning, quantization, retrieval-augmented generation, and adaptive computation policies.
Lack of Policy-Driven AI Cybersecurity Frameworks	Few systems enforce sustainability policies in AI-driven security [55].	Integrate ESG-aligned constraints (energy/carbon budgets) into AI design, training, and deployment, matching regulatory mandates.
Economic and Environmental Trade-off Blind Spots	Security gains are seldom weighed against cost and carbon footprint [23, 36].	Incorporate cost–benefit modeling that quantifies accuracy, latency, energy, emissions, and operational expenses.
Lack of Real-World Validation	Heavy reliance on simulation/testbed results limits practical relevance [59].	Conduct live deployments in smart cities, SCADA systems, and healthcare to assess robustness and sustainability in practice.
Ethical and Societal Risks in Sustainable Cybersecurity	Efficiency-driven designs risk bias, privacy breaches, and sustainability injustice [22, 57].	Create ethical guidelines, fairness-aware algorithms, and transparent auditing processes for both performance and sustainability.

dashboards combining explainability with sustainability metrics would strengthen transparency and accountability.

12.9. Inconsistent Model Adaptation to Resource Constraints

Adaptive modeling remains insufficient across existing studies. Although Gurram et al. [41] introduce energy-aware routing, most AI security models remain static and fail to respond to real-time changes in battery or CPU load. Meta-learning and RL-based resource feedback mechanisms can enhance adaptability, enabling systems to dynamically recalibrate their computational effort under fluctuating constraints. Such approaches would extend device lifetime while maintaining baseline protection, crucial for sustainable edge computing and mobile networks.

12.10. Absence of Multi-Objective Optimization Frameworks

Most research optimizes security or sustainability separately, missing the balance that can be achieved through multi-objective optimization [35]. Real-world systems require simultaneous trade-offs between accuracy, latency, and energy efficiency. Pareto optimization and evolutionary algorithms can achieve such a balance, ensuring robust yet energy-aware performance. Establishing standardized benchmarks that evaluate both security and sustainability dimensions will support the broader adoption of these frameworks.

12.11. Scarcity of Real-Time Energy-Aware AI Security Models

Runtime energy feedback is rarely integrated into AI-based cybersecurity systems. While Maiwada et al. [49]

propose efficient IDS for 5G, they lack real-time adaptation. Embedding energy sensors and developing adaptive scheduling mechanisms could allow models to self-tune energy use during operation. Designing standardized runtime APIs would promote interoperability and make dynamic energy optimization a core feature of sustainable AI security frameworks.

12.12. Environmental Cost of Model Retraining

Frequent retraining of large AI models significantly increases energy consumption [48]. Federated continual learning and adaptive retraining triggers can mitigate this impact. Incremental updates based on local threat patterns, rather than full retraining, preserve accuracy while reducing emissions. Such approaches align with sustainability thresholds and enhance both resilience and ecological efficiency in large-scale deployments.

12.13. Insufficient Cross-Layer Optimization

Cross-layer integration, spanning hardware, networking, and AI models, remains rare [61]. Efficiency gains achieved at one layer often counteract losses elsewhere. Joint optimization of communication protocols, model architectures, and hardware accelerators can maximize the synergy between security and sustainability. Automated orchestration frameworks capable of reallocating resources in real time based on both threat and energy conditions should be a focus for future research.

12.14. Underutilization of LLMs for Sustainability-Aware Security

LLMs are emerging as powerful tools for security automation but remain underexplored in sustainability contexts. Their high computational cost can be reduced through pruning, quantization, and retrieval-augmented generation. Lightweight LLMs tailored for energy-constrained environments could dynamically adjust their processing depth based on energy availability, ensuring both real-time adaptability and sustainable operation [22, 57].

12.15. Lack of Policy-Driven AI Cybersecurity Frameworks

Halabi et al. [55] highlight the gap between cybersecurity innovation and regulatory oversight. Few studies embed sustainability or energy budgets as enforceable constraints within AI models. Future frameworks should integrate ESG-aligned policies, ensuring models adhere to carbon and energy thresholds. Blockchain-enabled auditing and automated compliance monitoring could ensure transparency and accountability across AI-driven security deployments.

12.16. Economic and Environmental Trade-off Analysis

Economic and environmental cost modeling is seldom performed in cybersecurity research. Qiu et al. [36] show that FL can generate significantly higher CO_2 emissions than centralized models, while Jamshidi et al. [23] demonstrate trade-offs between performance and energy. Integrating cost-benefit and lifecycle impact analyses would enable stakeholders to assess the sustainability benefits of AI security systems in operational settings.

12.17. Lack of Real-World Validation

Most AI-based sustainability-aware security systems remain limited to simulations. Practical deployment, such as in Fraga-Lamas et al. [59], is rare. Real-world pilots in smart cities, SCADA systems, and healthcare networks are crucial for testing scalability, robustness, and actual energy performance. Collaborative field trials among academia, industry, and governments could help close this validation gap and accelerate the practical adoption of these technologies.

12.18. Ethical and Societal Implications

Sustainability-focused cybersecurity also introduces ethical challenges. Lightweight detection models may compromise fairness, while extensive data collection raises privacy issues [22, 57]. Incorporating fairness-aware training, differential privacy, and transparent auditing can address these concerns. Engaging diverse stakeholders in the design process ensures that future AI-driven cybersecurity aligns with principles of equity, privacy, and environmental justice.

12.19. Emerging AI Paradigms in Sustainable Cybersecurity

The future of sustainable cybersecurity will be shaped by emerging AI paradigms that extend beyond current ML, DL, and FL approaches. LLMs are expected to support autonomous threat reasoning, adaptive policy generation, and explainable security analysis, provided they can be fine-tuned for low-power, domain-specific deployments. Similarly, spiking neural networks offer neuromorphic efficiency, enabling real-time anomaly detection with minimal energy use in edge environments. Quantum AI, though in its infancy, presents opportunities for ultra-secure cryptographic mechanisms and accelerated model optimization, potentially transforming the scalability and sustainability of cybersecurity. These paradigms collectively represent the next frontier, where computational intelligence and ecological responsibility converge to enable resilient, green, and intelligent cyber defense systems.

13. Conclusion

This study emphasizes the crucial importance of incorporating sustainability considerations into AI-driven cybersecurity, particularly within smart and interconnected infrastructures. While advanced AI techniques, such as DL and FL, have significantly enhanced security intelligence and system resilience, and blockchain technologies have improved trust and data integrity, sustainability factors, including energy efficiency, lifecycle carbon emissions, and environmental cost, are often treated as secondary objectives. Through a systematic review of the existing literature, this

paper identifies key technical, environmental, and policy-related challenges, revealing a persistent imbalance between cybersecurity performance and sustainable operation. To address this gap, the study presents a unified taxonomy of AI-based cybersecurity approaches and synthesizes current practices through a structured analysis of sustainability objectives, evaluation metrics, and application domains. In addition, a conceptual framework is introduced as part of the research agenda, linking energy-aware design principles, lifecycle carbon accounting, and adaptive intelligence to support the development of sustainability-aware security. Based on these insights, a prioritized research agenda is outlined, emphasizing the need for standardized sustainability metrics, cross-layer and multi-objective optimization frameworks, lightweight and energy-efficient AI architectures, real-world validation of sustainable cybersecurity solutions, and the integration of policy, ethical, and governance mechanisms. Collectively, these directions provide a concrete roadmap toward secure, sustainable, and intelligent digital infrastructures, ensuring that future cybersecurity advancements align technological innovation with global environmental and sustainability objectives.

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